

The Pol-InSAR Course

Synthetic Aperture Radar (SAR)

Part 6: SAR Polarimetry

Prepared by DLR-HR's Pol-InSAR Team

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ESAMAAP

EEBI  MASS

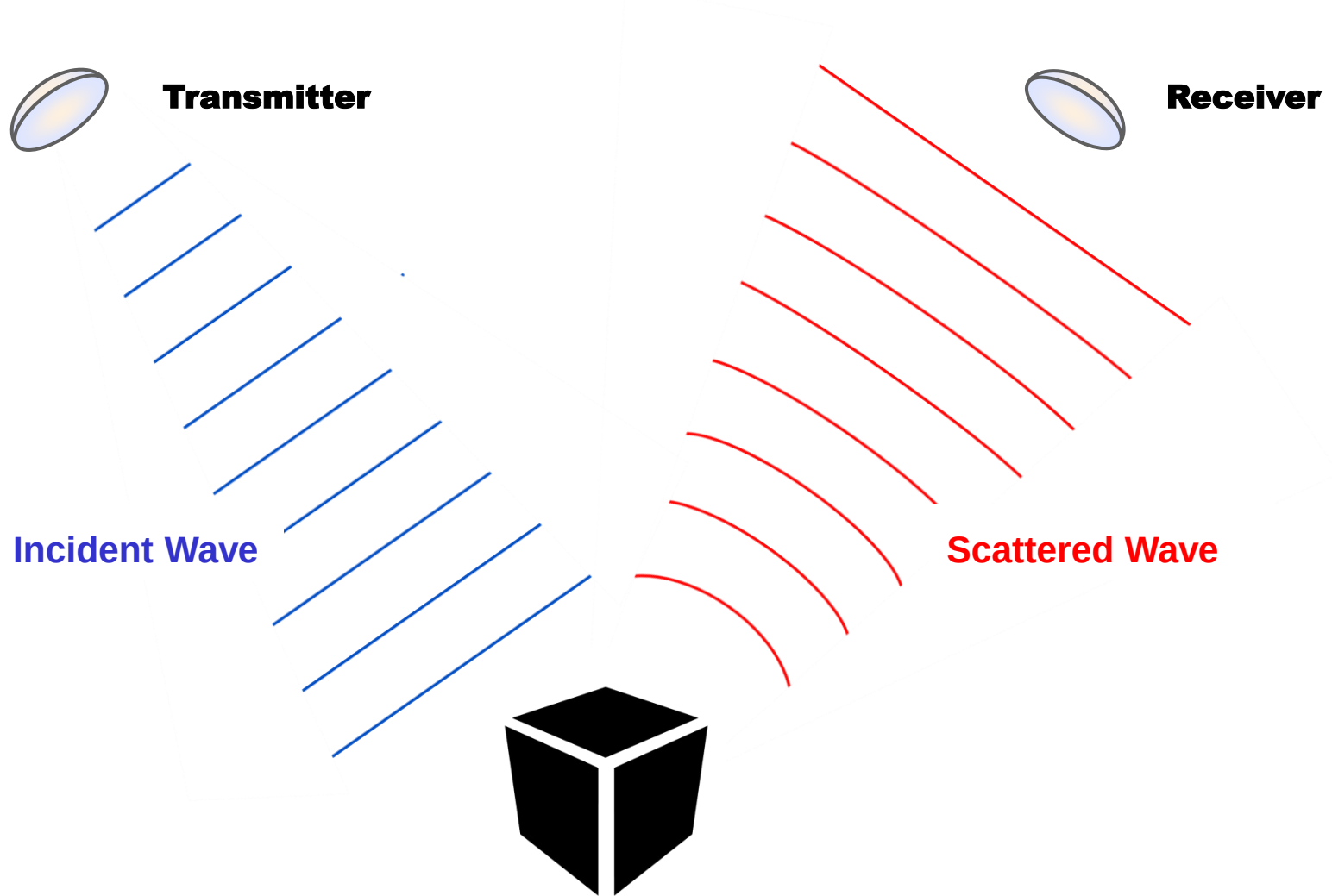
F-SAR (DLR), Kaufbeuren, X-Band, HH - Pol



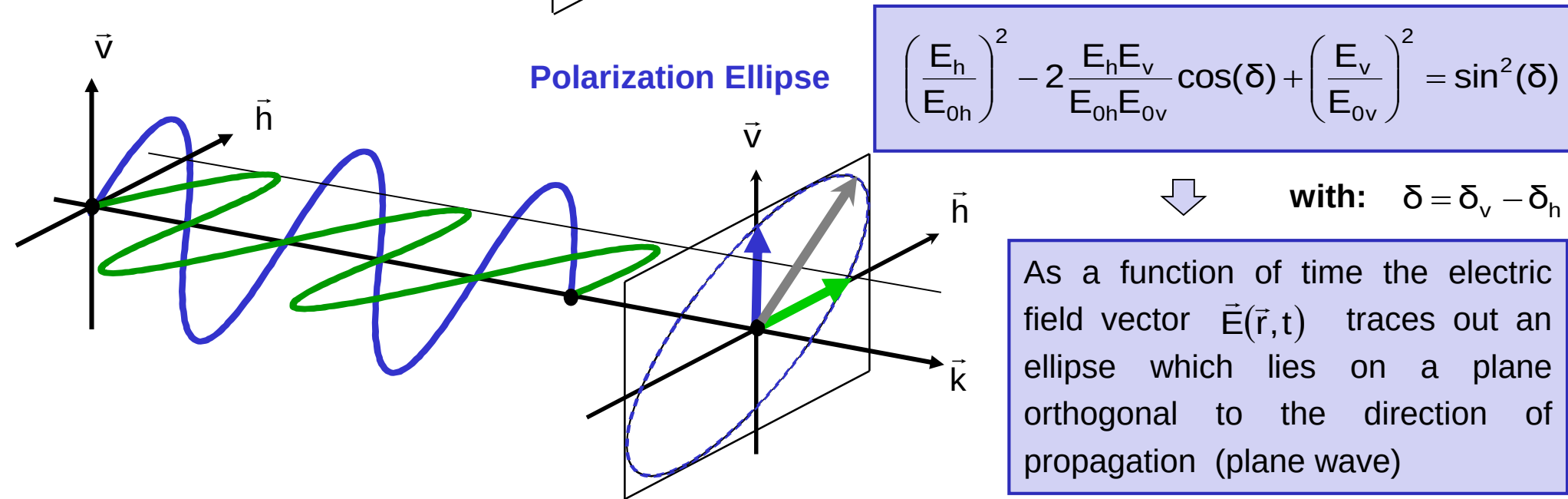
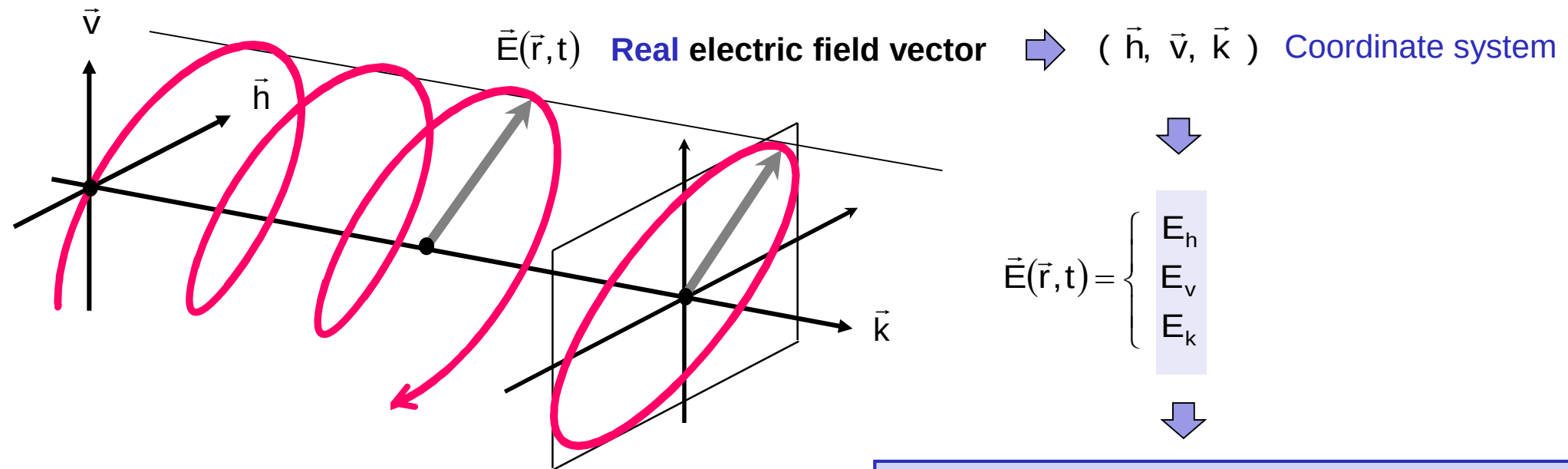
F-SAR (DLR), Kaufbeuren, X-Band, Quad-Pol



The (Inverse) Scattering Problem

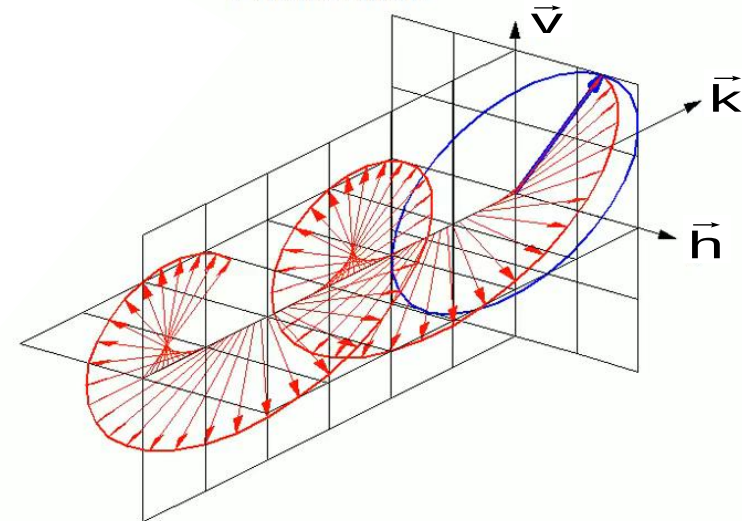


The Scatterer transforms the incident into the scattered wave

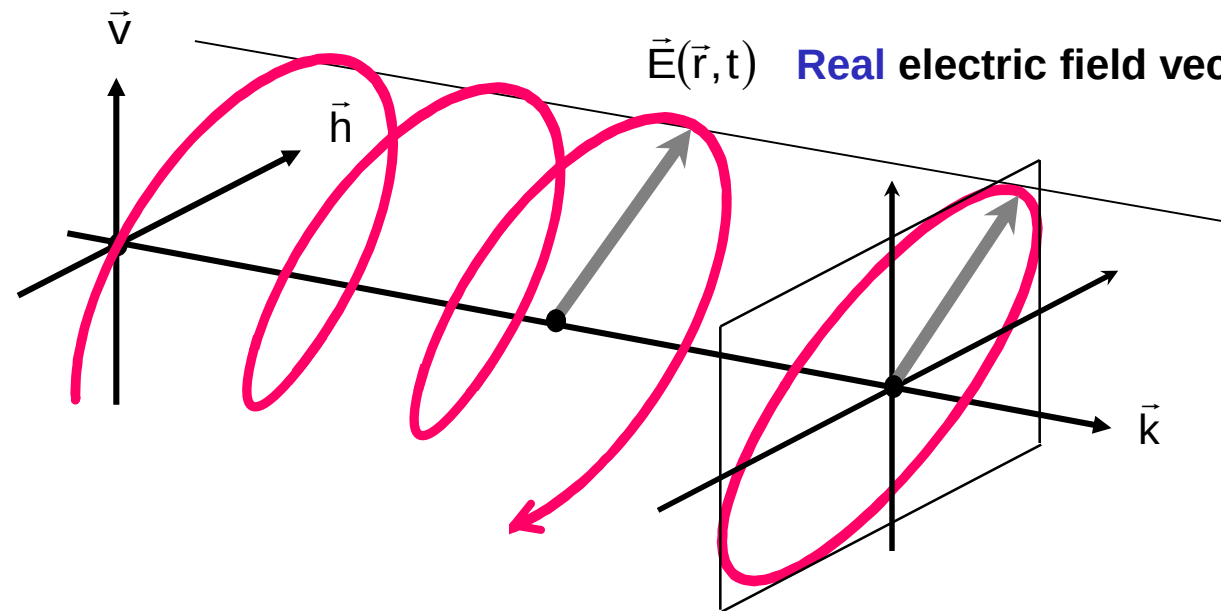


Polarisation is an intrinsic property of the wave; independent of the presence of a coordinate system. But, a coordinate system is needed to describe the polarisation state of the wave.

Elliptical (Right Hand)
Polarization

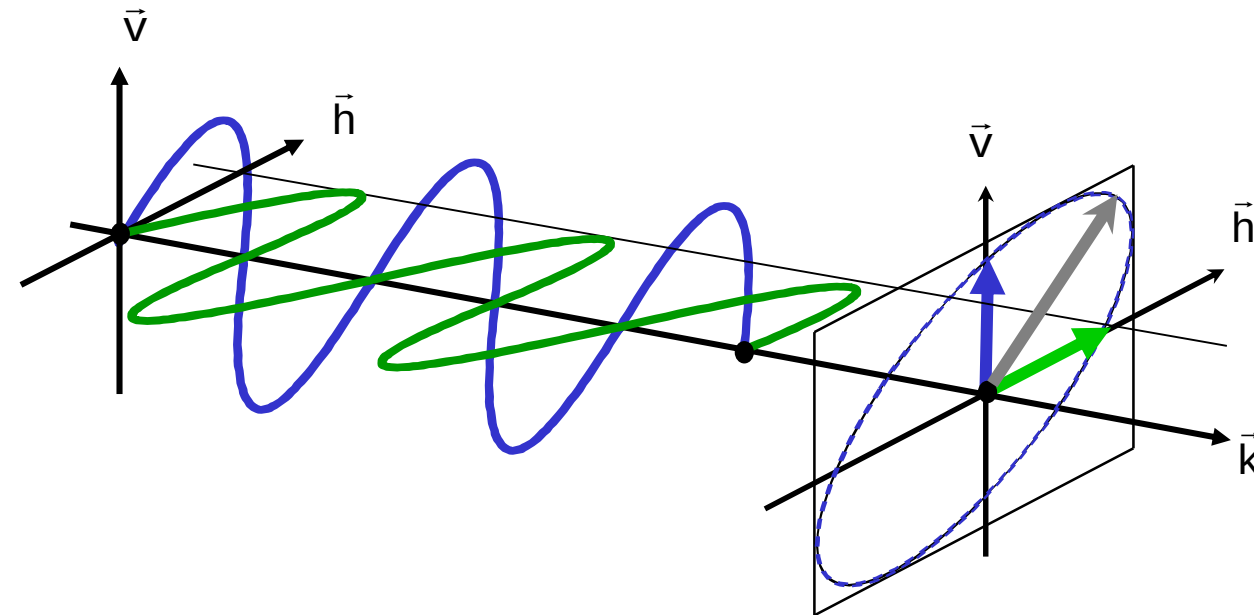


$$\vec{E}(\vec{r}, t) = \begin{cases} E_h = E_{0h} \cos(\omega t - kr - \delta_h) \\ E_v = E_{0v} \cos(\omega t - kr - \delta_v) \\ E_k = 0 \end{cases} \rightarrow \begin{bmatrix} E_{0h} \cos(\omega t - kr - \delta_h) \\ E_{0v} \cos(\omega t - kr - \delta_v) \end{bmatrix} \rightarrow \begin{bmatrix} E_{0h} \exp(i\delta_h) \\ E_{0v} \exp(i\delta_v) \end{bmatrix} \exp(i\omega t - ikr)$$



$\Downarrow$ **Plane Harmonic Wave**

$$\vec{E}(\vec{r}, t) = \begin{cases} E_h = E_{0h} \cos(\omega t - kr - \delta_h) \\ E_v = E_{0v} \cos(\omega t - kr - \delta_v) \\ E_k = 0 \end{cases}$$

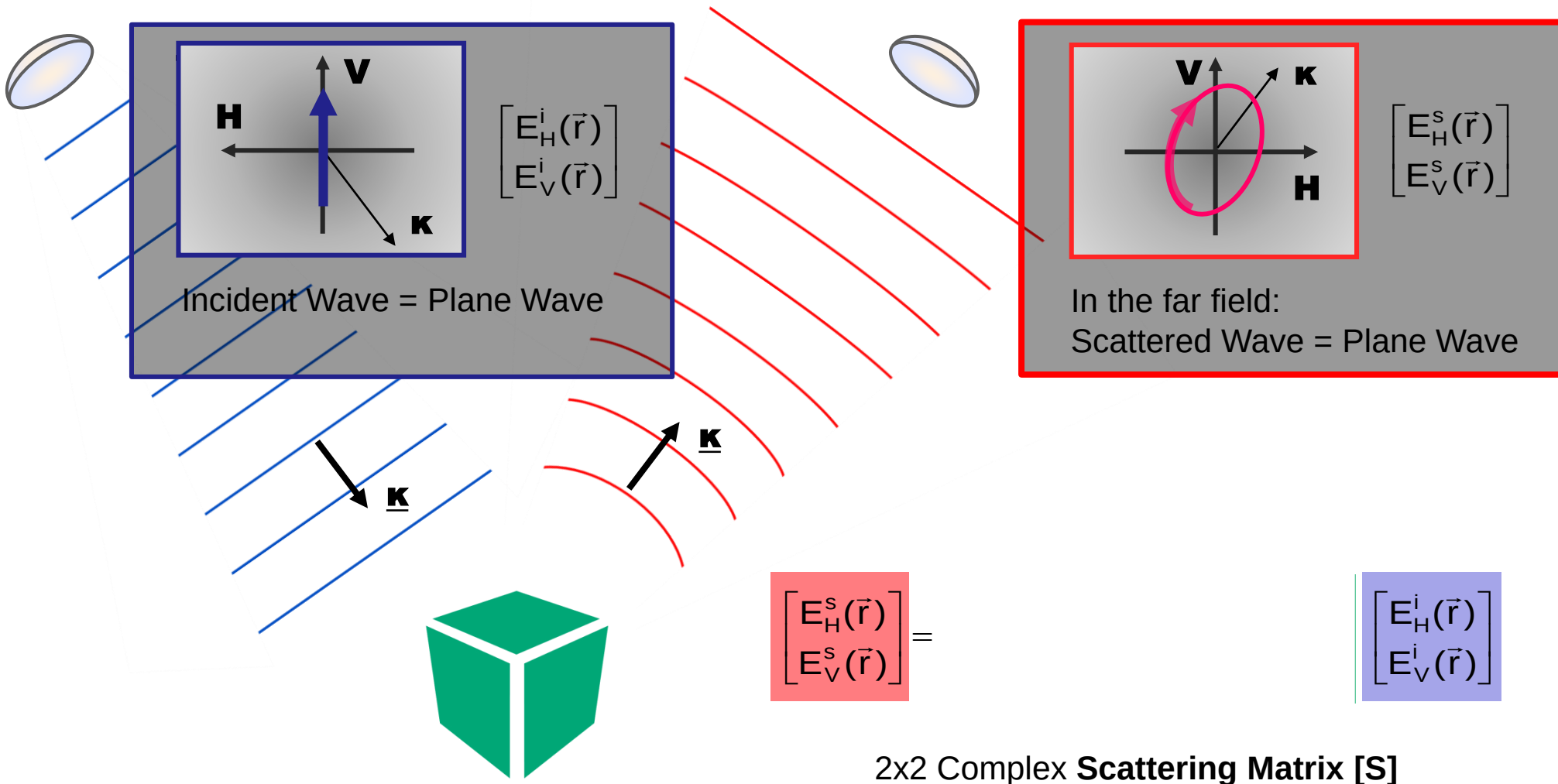


Jones vector (polarization descriptor):

$$\vec{E} = \begin{bmatrix} E_h \\ E_v \end{bmatrix} = \begin{bmatrix} E_{0h} \exp(i\delta_h) \\ E_{0v} \exp(i\delta_v) \end{bmatrix} \in \mathbb{C}_2$$

The description can be considered as a component representation in the 2D complex vector space with basis $(\vec{h}, \vec{v})$

The Polarimetric Scattering Problem



The Scatterer changes the polarisation of the incident wave.

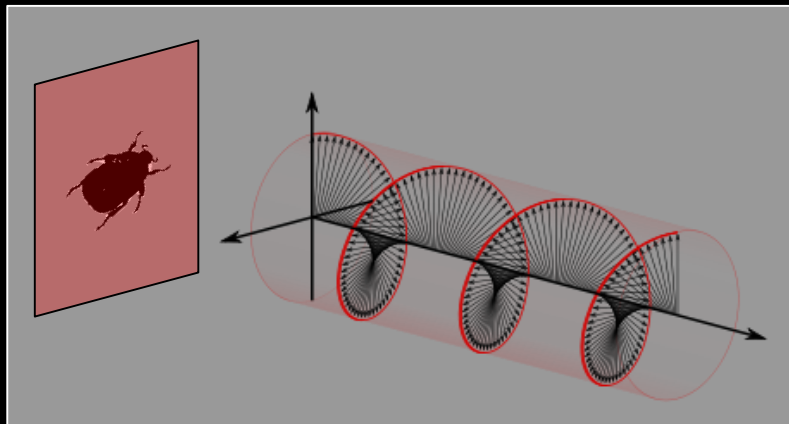
In the literature also the $[S] = \frac{\exp(ikr)}{ikr} \begin{bmatrix} S_{HH}(\vec{r}) & S_{HV}(\vec{r}) \\ S_{VH}(\vec{r}) & S_{VV}(\vec{r}) \end{bmatrix}$ definition can be found. Be aware that there are indeed slight differences in the definition of $[S]$.

Chirality-induced polarization effects in the cuticle of scarab beetles: 100 years after Michelson

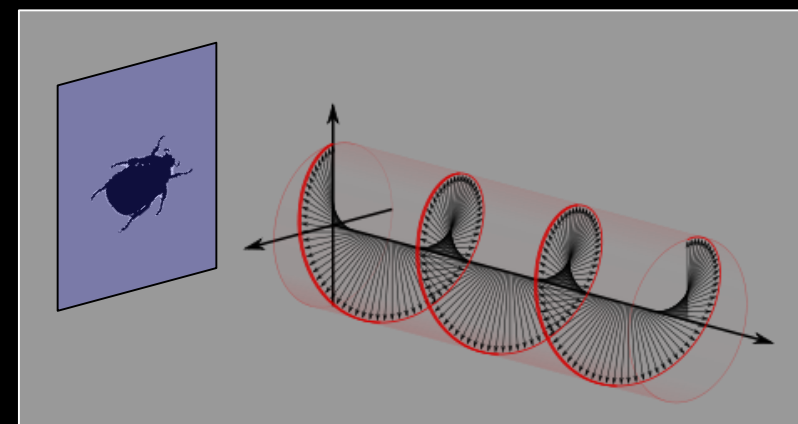
Hans Arwin*, Roger Magnusson, Jan Landin and Kenneth Järrendahl



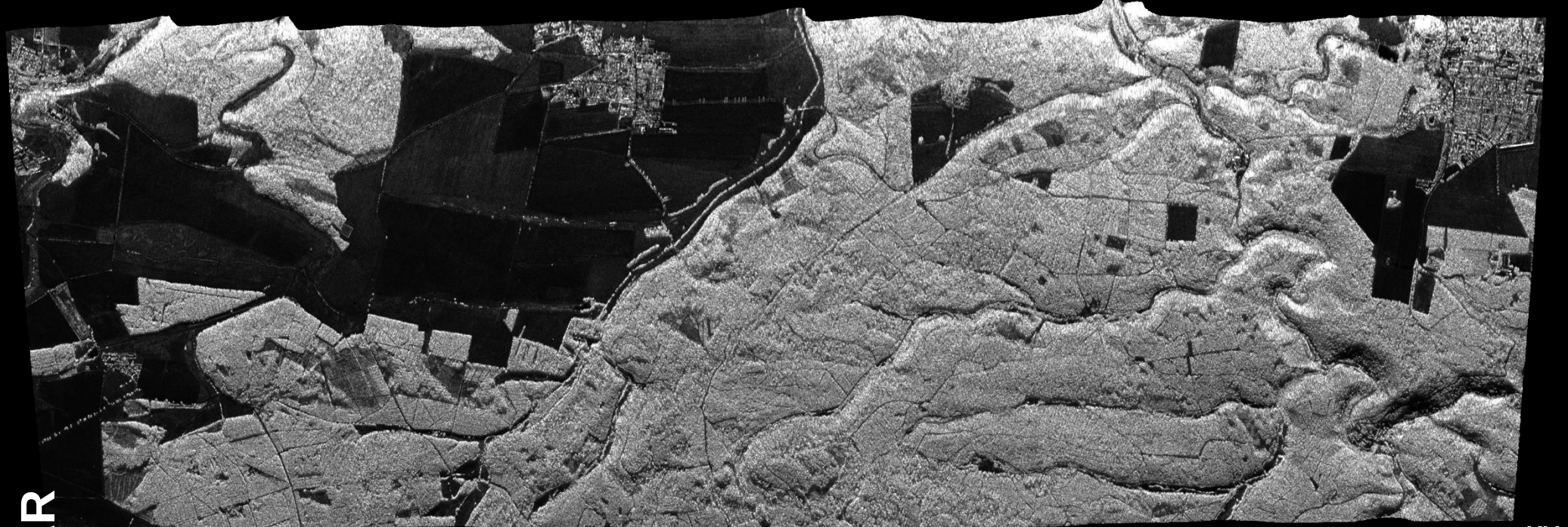
Cetonia Aurata seen through a left circular filter



Cetonia Aurata seen through a right circular filter



Polarimetric L-band SAR



HV

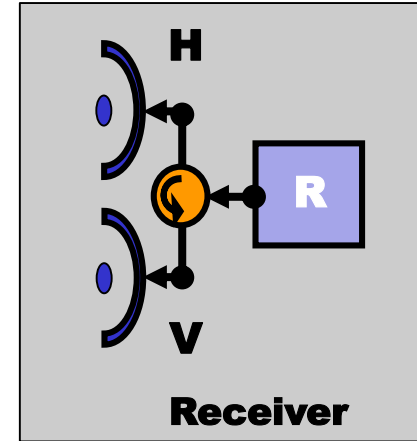
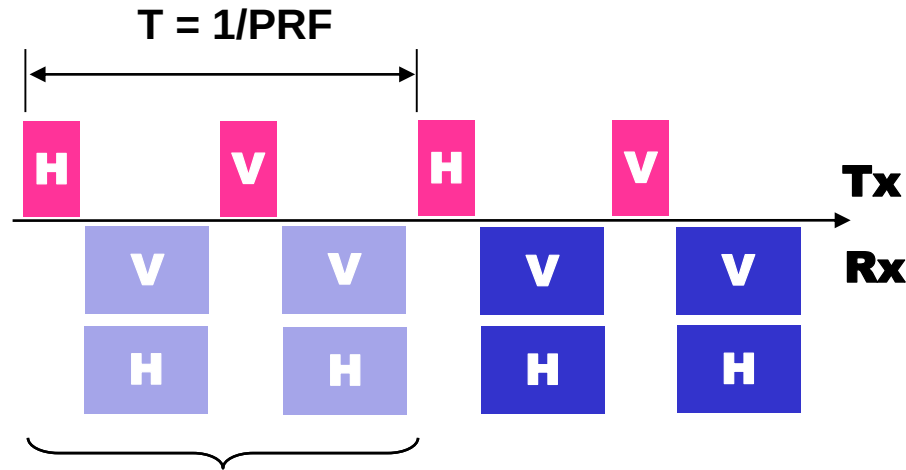
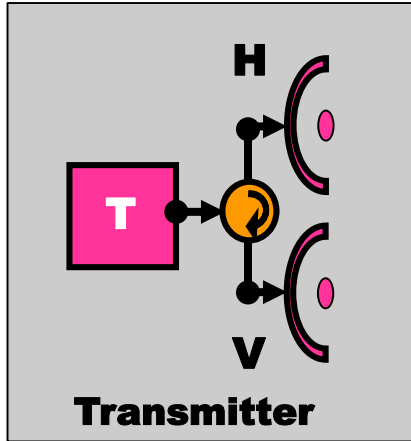


VV

E-SAR, SOIMEX 13, Bode Test site, L-band

Bi- & Mono-Static Measurement of the Scattering Matrix

Bistatic
T & R Separated



H $\begin{bmatrix} E_H^T \\ E_V^T \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

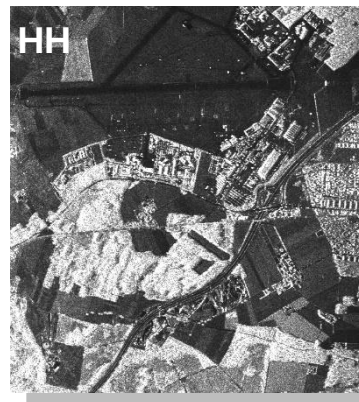
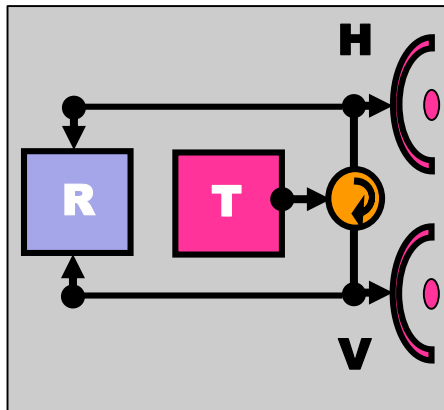
V $\begin{bmatrix} E_H^T \\ E_V^T \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

H $\begin{bmatrix} E_H^R \\ E_V^R \end{bmatrix} = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} \begin{bmatrix} E_H^T \\ E_V^T \end{bmatrix} = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} S_{HH} \\ S_{VH} \end{bmatrix}$

V $\begin{bmatrix} E_H^R \\ E_V^R \end{bmatrix} = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} \begin{bmatrix} E_H^T \\ E_V^T \end{bmatrix} = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} S_{HV} \\ S_{VV} \end{bmatrix}$

$\begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix}$

Monostatic
T & R Colocated



Scattering Amplitudes Images

HH

HV



VH

VV

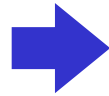
Azimuth
Range



Scattering Matrix – Number of Independent Parameters

... also known as the Jones Matrix in the bistatic and Sinclair Matrix in the monostatic case

$$[S] = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix}$$



4 Complex Scattering Amplitudes:

$$S_{IJ} = |S_{IJ}| \exp(i\phi_{IJ}) = f(\text{Frequency, Scattering, Geometry})$$

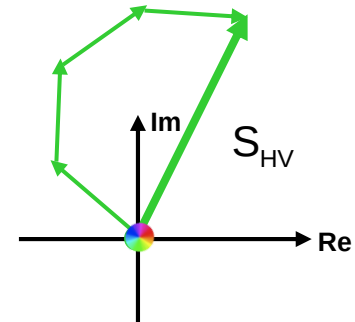
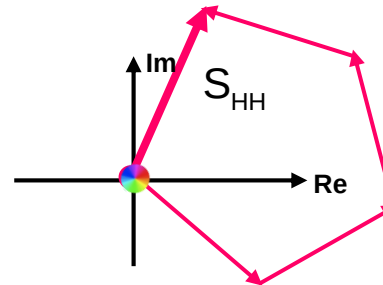
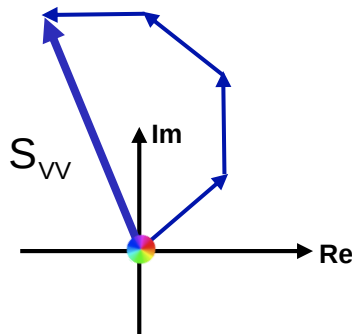
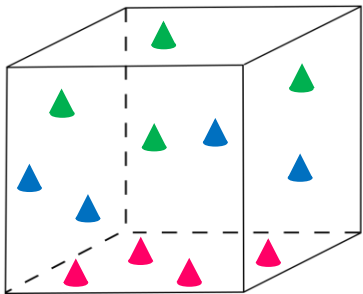
Total Scattered Power: $TP = \text{Span}([S]) = \text{Trace}([S][S]^+) = |S_{HH}|^2 + |S_{HV}|^2 + |S_{VH}|^2 + |S_{VV}|^2$

Bistatic Scattering Matrix: $S_{HV} \neq S_{VH}$

Absolute Phase Factor

$$[S] = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} = \begin{bmatrix} |S_{HH}| \exp(i\phi_{HH}) & |S_{HV}| \exp(i\phi_{HV}) \\ |S_{VH}| \exp(i\phi_{VH}) & |S_{VV}| \exp(i\phi_{VV}) \end{bmatrix} = e^{i\phi_{VV}} \begin{bmatrix} |S_{HH}| \exp(i(\phi_{HH} - \phi_{VV})) & |S_{HV}| \exp(i(\phi_{HV} - \phi_{VV})) \\ |S_{VH}| \exp(i(\phi_{VH} - \phi_{VV})) & |S_{VV}| \end{bmatrix}$$

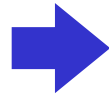
The bistatic scattering matrix contains **seven** independent parameters: 4 Amplitudes & 3 Phases



Scattering Matrix

... also known as the Jones Matrix in the bistatic and Sinclair Matrix in the monostatic case

$$[S] = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix}$$



4 Complex Scattering Amplitudes:

$$S_{IJ} = |S_{IJ}| \exp(i\varphi_{IJ}) = f(\text{Frequency, Scattering, Geometry})$$

Total Scattered Power: $TP = \text{Span}([S]) = \text{Trace}([S][S]^+) = |S_{HH}|^2 + |S_{HV}|^2 + |S_{VH}|^2 + |S_{VV}|^2$

Bistatic Scattering Matrix: $S_{HV} \neq S_{VH}$

Absolute Phase Factor

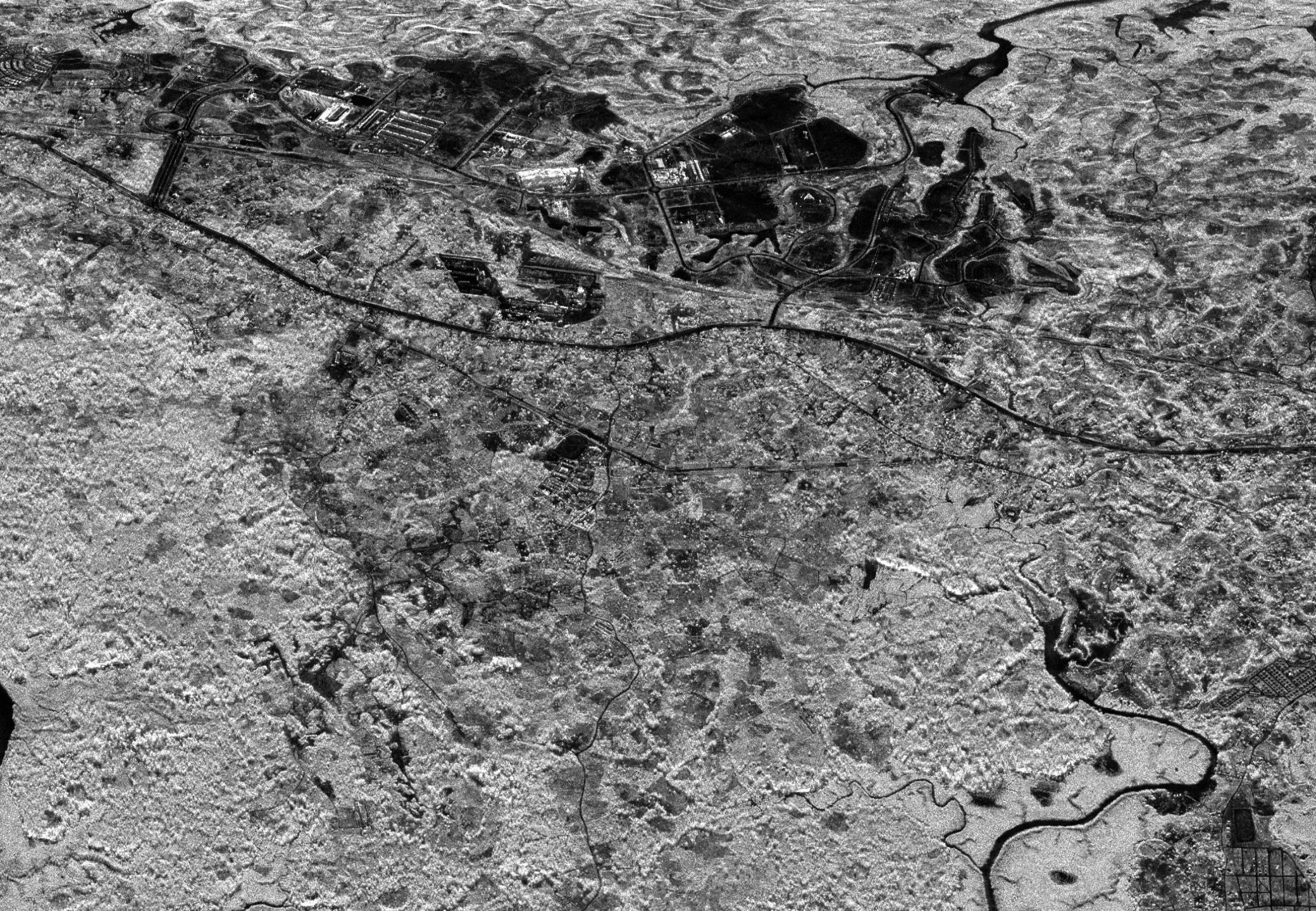
$$[S] = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} = \begin{bmatrix} |S_{HH}| \exp(i\varphi_{HH}) & |S_{HV}| \exp(i\varphi_{HV}) \\ |S_{VH}| \exp(i\varphi_{VH}) & |S_{VV}| \exp(i\varphi_{VV}) \end{bmatrix} = e^{i\varphi_{VV}} \begin{bmatrix} |S_{HH}| \exp(i(\varphi_{HH} - \varphi_{VV})) & |S_{HV}| \exp(i(\varphi_{HV} - \varphi_{VV})) \\ |S_{VH}| \exp(i(\varphi_{VH} - \varphi_{VV})) & |S_{VV}| \end{bmatrix}$$

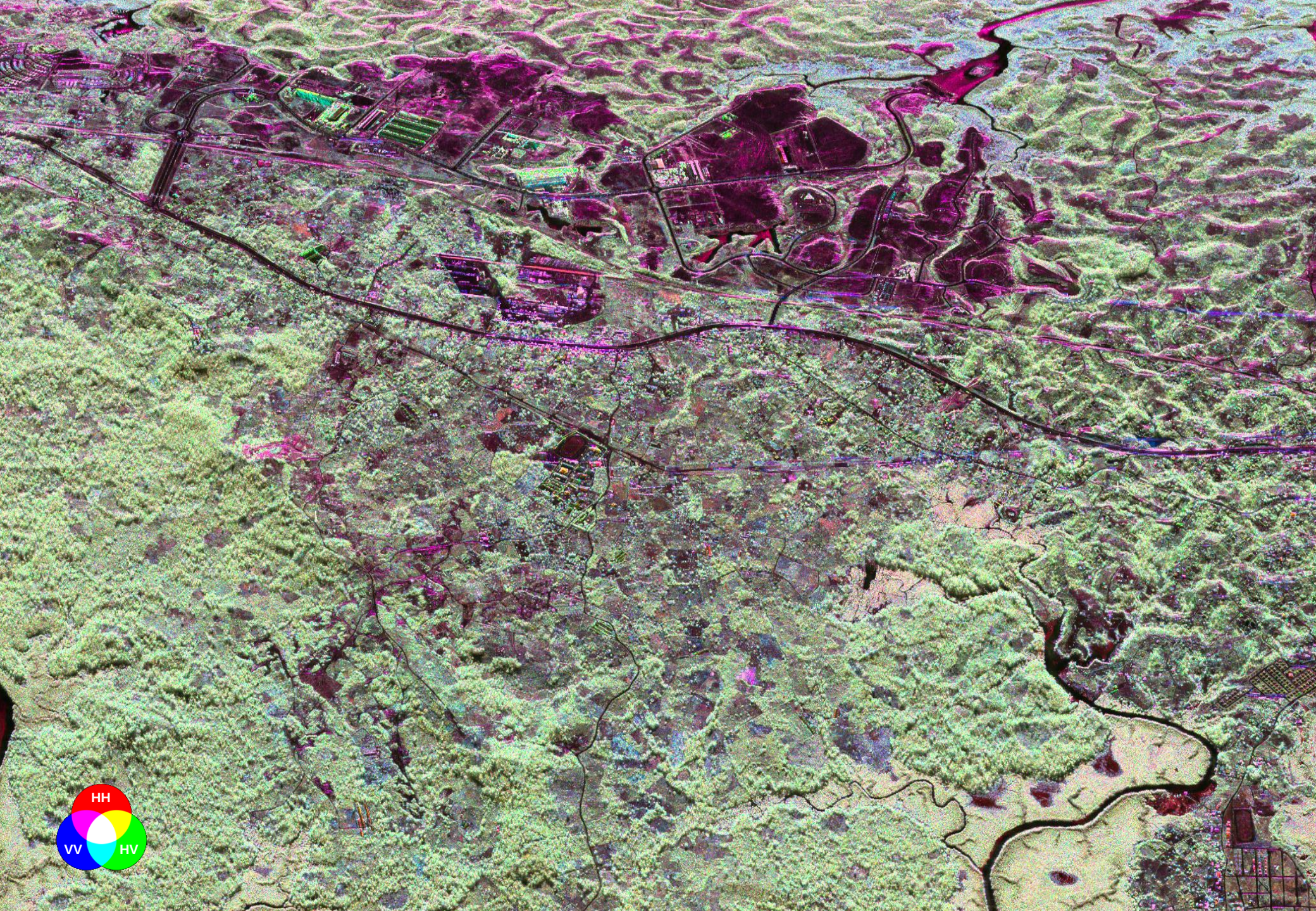
The bistatic scattering matrix contains **seven** independent parameters: 4 Amplitudes & 3 Phases

Monostatic Scattering Matrix: $S_{HV} = S_{VH} = S_{XX}$

$$[S] = \begin{bmatrix} S_{HH} & S_{XX} \\ S_{XX} & S_{VV} \end{bmatrix} = \begin{bmatrix} |S_{HH}| \exp(i\varphi_{HH}) & |S_{XX}| \exp(i\varphi_{XX}) \\ |S_{XX}| \exp(i\varphi_{XX}) & |S_{VV}| \exp(i\varphi_{VV}) \end{bmatrix} = e^{i\varphi_{VV}} \begin{bmatrix} |S_{HH}| \exp(i(\varphi_{HH} - \varphi_{VV})) & |S_{XX}| \exp(i(\varphi_{XX} - \varphi_{VV})) \\ |S_{XX}| \exp(i(\varphi_{XX} - \varphi_{VV})) & |S_{VV}| \end{bmatrix}$$

The monostatic scattering matrix contains **five** independent parameters: 3 Amplitudes & 2 Phases



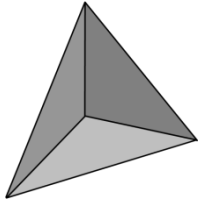


Scattering Matrices of Canonical Scatterers



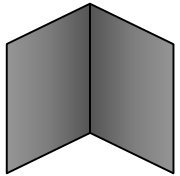
Sphere

$$[S] = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$



Trihedral

$$[S] = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$



Dihedral

$$[S] = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$



V Dipol

$$[S] = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$



H Dipol

$$[S] = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$



L Helix

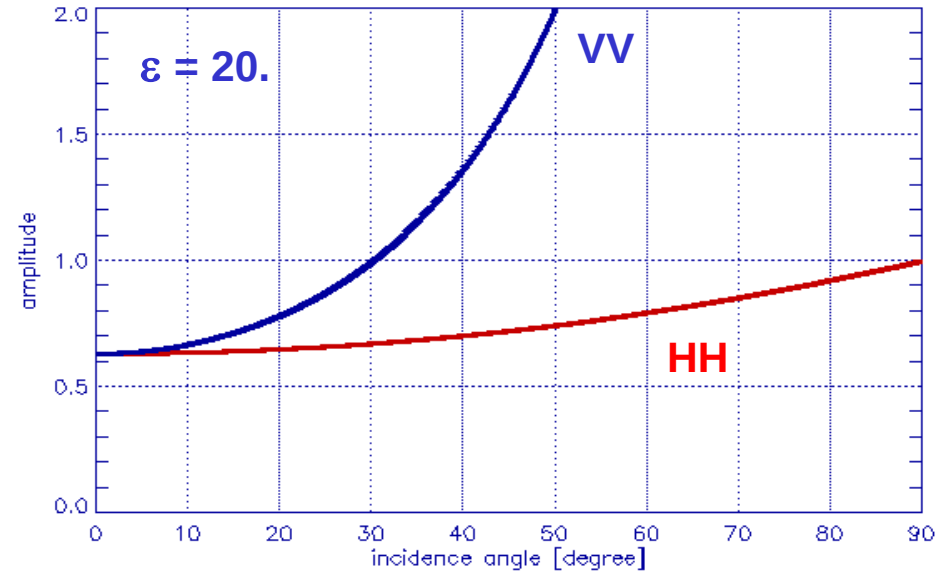
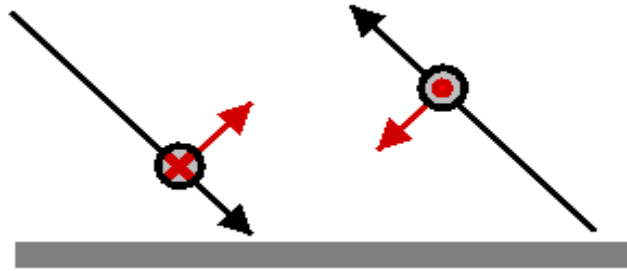
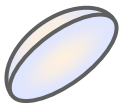
$$[S] = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} = \begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix}$$



R Helix

$$[S] = \begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix}$$

Canonical Scattering Processes: Bragg Scattering



Scattering Matrix: $[S] = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} = \begin{bmatrix} R_H & 0 \\ 0 & R_V \end{bmatrix}$

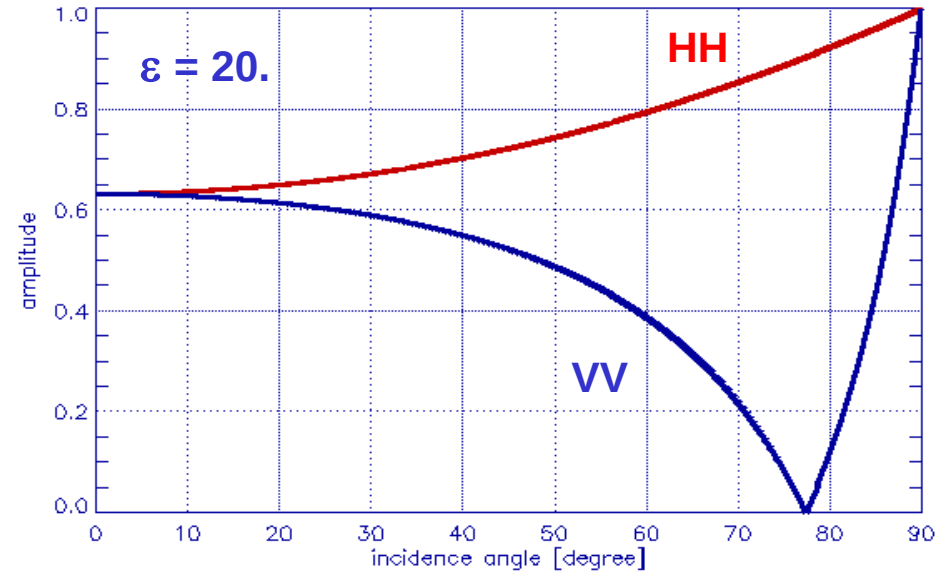
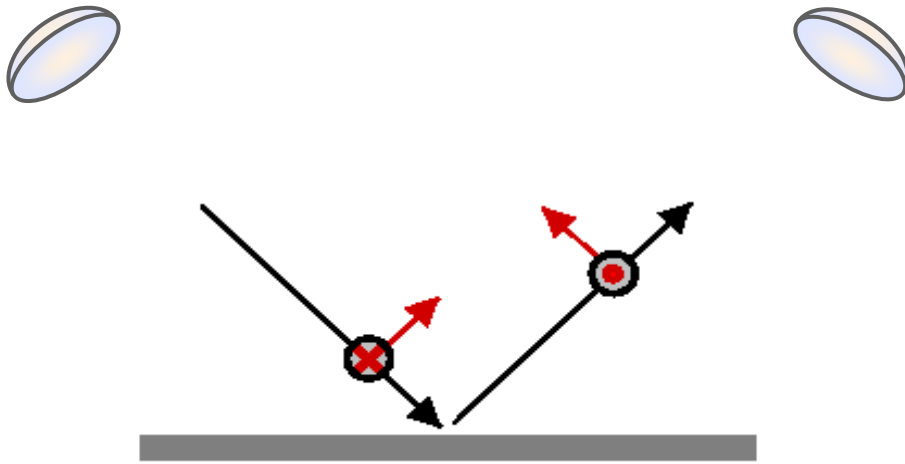
Bragg Scattering Coefficients: $R_H = \frac{\cos \theta - \sqrt{\epsilon - \sin^2 \theta}}{\cos \theta + \sqrt{\epsilon - \sin^2 \theta}}$ and $R_V = \frac{(\epsilon - 1)[\sin^2 \theta - \epsilon(1 + \sin^2 \theta)]}{\epsilon \cos \theta + \sqrt{\epsilon - \sin^2 \theta}}$

... which depend on the incidence angle θ and the dielectric constant of the surface ϵ

► Soil moisture inversion applications !

Bragg Characteristics: S_{HH} and S_{VV} are in phase and $|S_{HH}| \leq |S_{VV}|$

Canonical Scattering Processes: Fresnel Reflection



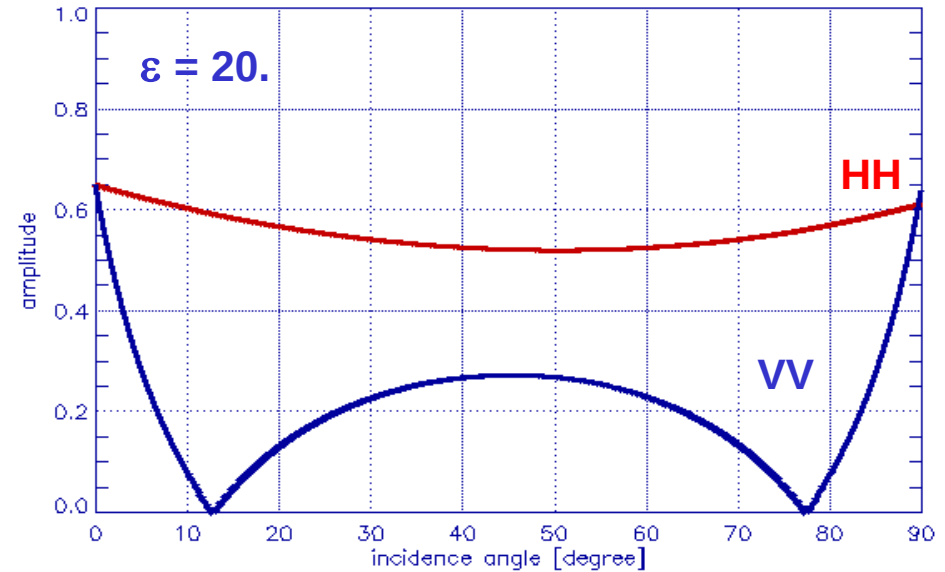
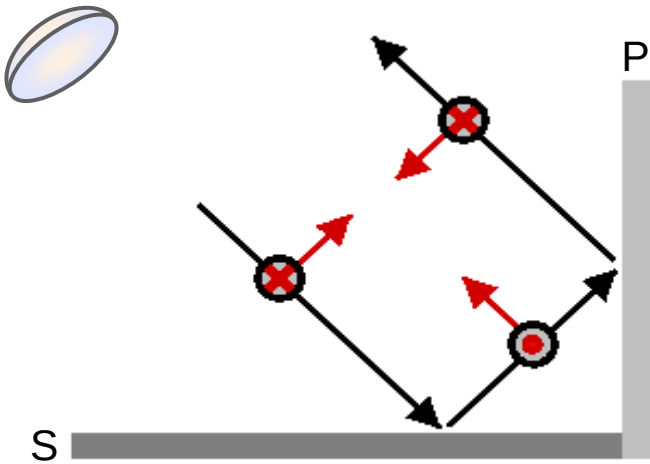
Scattering Matrix: $[S] = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} = \begin{bmatrix} R_H & 0 \\ 0 & R_V \end{bmatrix}$

Fresnel Reflection Coefficients: $R_H = \frac{\cos \theta - \sqrt{\epsilon - \sin^2 \theta}}{\cos \theta + \sqrt{\epsilon - \sin^2 \theta}}$ and $R_V = \frac{\epsilon \cos \theta - \sqrt{\epsilon - \sin^2 \theta}}{\epsilon \cos \theta + \sqrt{\epsilon - \sin^2 \theta}}$

... which depend on the incidence angle θ and the dielectric constant of the surface ϵ

Fresnel Characteristics: S_{HH} and S_{VV} are in phase and $|S_{HH}| \geq |S_{VV}|$

Canonical Scattering Processes: Dihedral Scattering



$$\text{Scattering Matrix: } [S] = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & e^i \end{bmatrix} \begin{bmatrix} R_{SS} & 0 \\ 0 & R_{PS} \end{bmatrix} \begin{bmatrix} R_{ST} & 0 \\ 0 & R_{PT} \end{bmatrix} = \begin{bmatrix} R_{SS}R_{ST} & 0 \\ 0 & -R_{PS}R_{PT}e^i \end{bmatrix}$$

Fresnel	Horizontal Plane	$R_{SS} = \frac{\cos \theta - \sqrt{\epsilon_S - \sin^2 \theta}}{\cos \theta + \sqrt{\epsilon_S - \sin^2 \theta}}$	and	$R_{PS} = \frac{\epsilon_S \cos \theta - \sqrt{\epsilon_S - \sin^2 \theta}}{\epsilon_S \cos \theta + \sqrt{\epsilon_S - \sin^2 \theta}}$
Reflection				
Coefficients	Vertical Plane	$R_{ST} = \frac{\cos \theta' - \sqrt{\epsilon_P - \sin^2 \theta'}}{\cos \theta' + \sqrt{\epsilon_P - \sin^2 \theta'}}$	and	$R_{PT} = \frac{\epsilon_P \cos \theta' - \sqrt{\epsilon_P - \sin^2 \theta'}}{\epsilon_P \cos \theta' + \sqrt{\epsilon_P - \sin^2 \theta'}}$

Dihedral Characteristics: S_{HH} and S_{VV} are phase shifted by 180° (i.e., π) and $|S_{HH}| \geq |S_{VV}|$

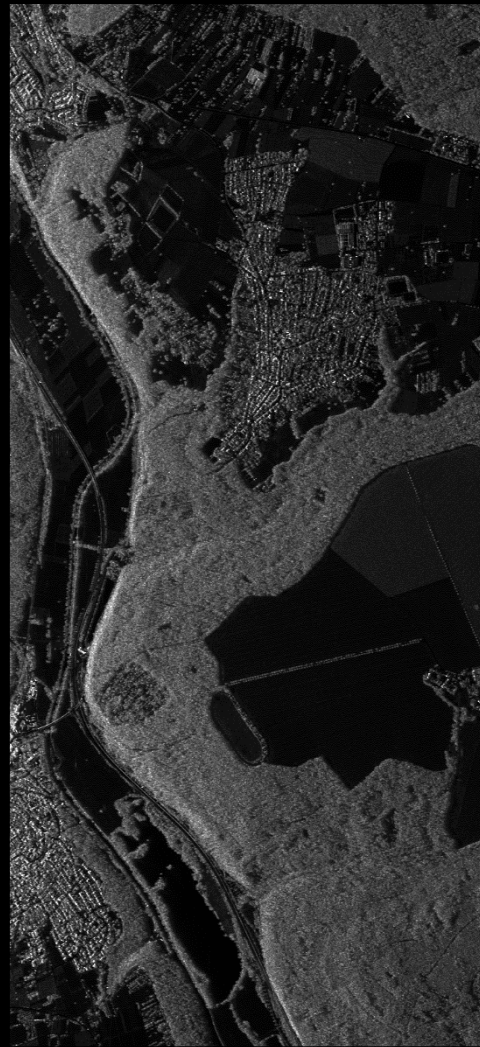
Scattering Amplitudes



HH



HV



VH



VV

Scattering Amplitudes



HH



HV

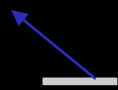


VH

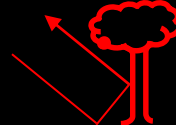


VV

Surface Scattering: $|VV| > |HH|$



$|HV|=0$

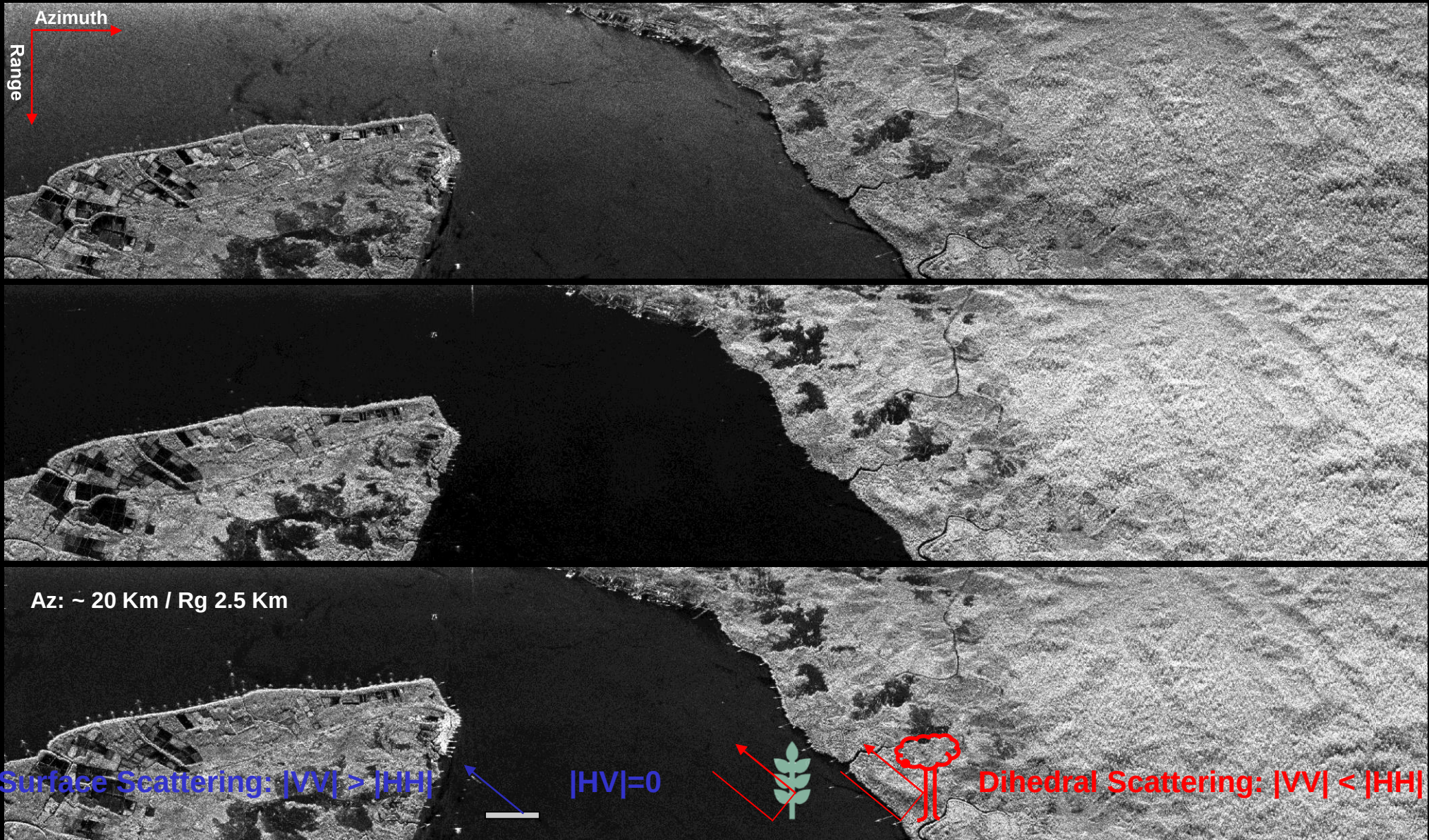


Dihedral Scattering: $|VV| < |HH|$



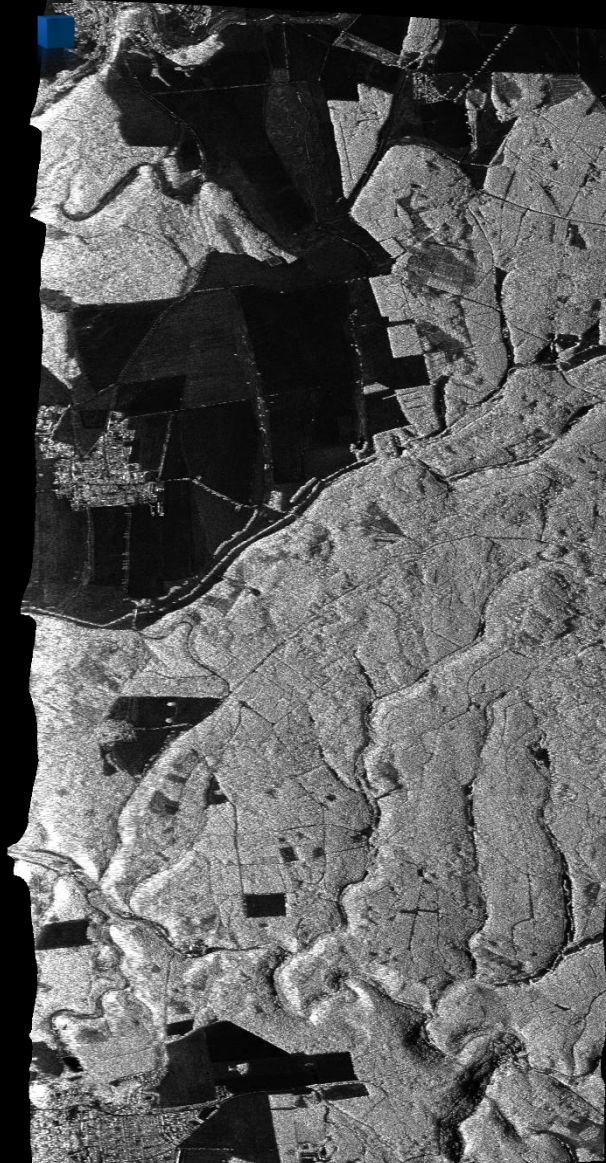
Polarimetric Backscatter Images: HH, HV, VV. Which is which ?

E-SAR L-band / Test Site: Sungai Wain South, Indonesia





Polarimetric Backscatter Images: HH, HV, VV. Which is which ?



Surface Scattering: $|VV| > |HH|$



$|HV|=0$



Dihedral Scattering: $|VV| < |HH|$

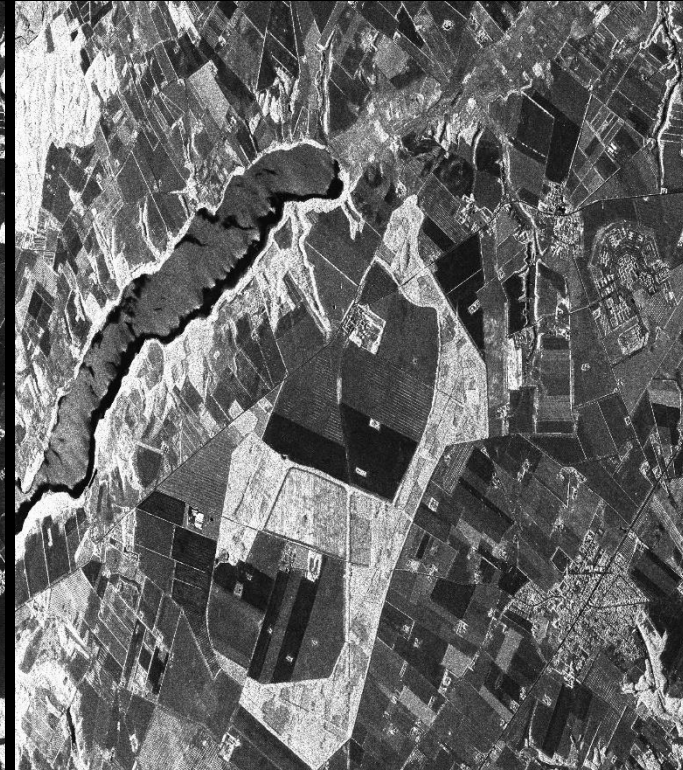
E-SAR, SOIMEX 13, Bode Test site, L-band



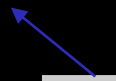
Polarimetric Backscatter Images: HH, HV, VV. Which is which ?

L-band

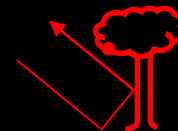
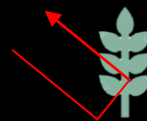
EMI-SAR / Test Site: Foulum, Denmark



Surface Scattering: $|VV| > |HH|$



$|HV|=0$



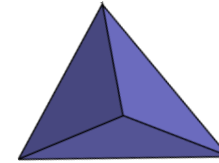
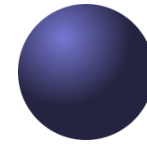
Dihedral Scattering: $|VV| < |HH|$

Pauli Decomposition

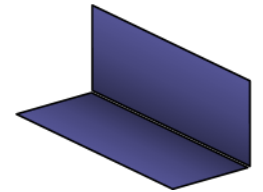
... decomposition of the scattering matrix into four scattering mechanisms:

$$[S] = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} = \begin{bmatrix} a+b & c-id \\ c+id & a-b \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + b \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} + c \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + d \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

- $[S_a] = a \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ **Single Scattering:** $a = S_{HH} + S_{VV}$

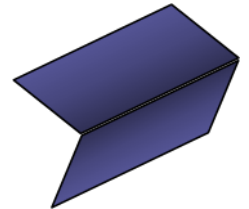


- $[S_b] = b \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ **Dihedral Scattering:** $b = S_{HH} - S_{VV}$



- $[S_c] = c \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ **Dihedral Scattering:** $c = S_{HV} + S_{VH}$

(... rotated by $\pi/4$ about the LOS – Line of Sight)



- $[S_d] = d \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$ **Helix Scattering:** $d = S_{HV} - S_{VH} = 0$ in backscattering !!!



In the monostatic case $[S]$ is decomposed into three orthogonal scattering mechanisms $[S_a]$, $[S_b]$, $[S_c]$.

Scattering Amplitudes



HH



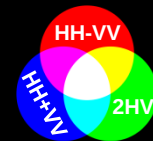
HV



VV



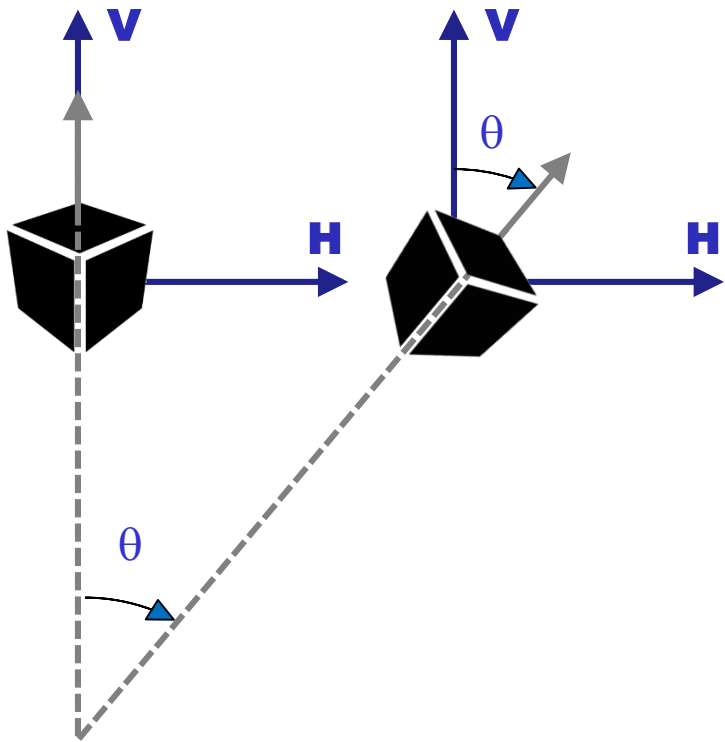
Pauli RGB



Line of Sight (LoS) Rotations

LOS Rotation of the Scatterer by θ = LOS Rotation of the Antenna by $-\theta$

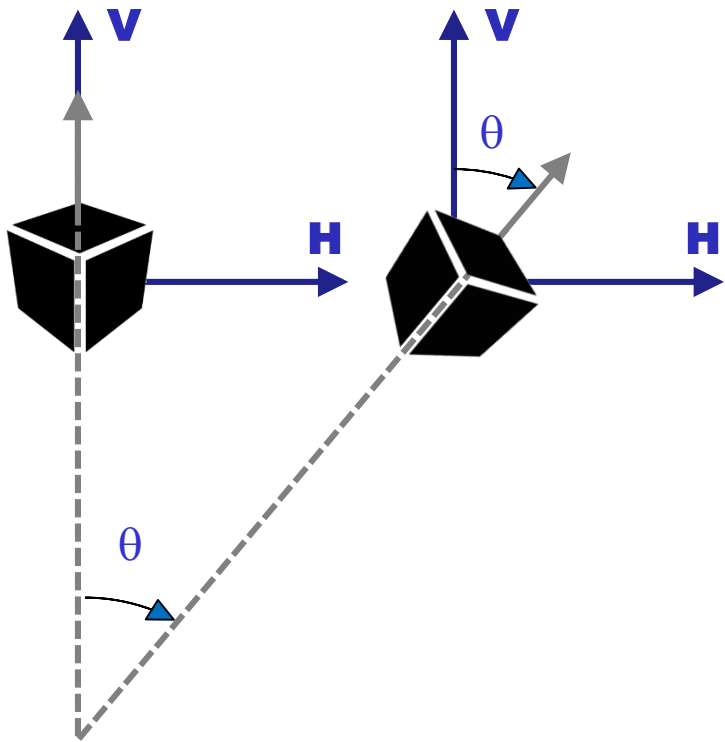
$$[S] = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} \quad \rightarrow \quad [S(\theta)] = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$



Line of Sight (LoS) Rotations

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$$[S] = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} \quad \rightarrow \quad [S(\theta)] = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$



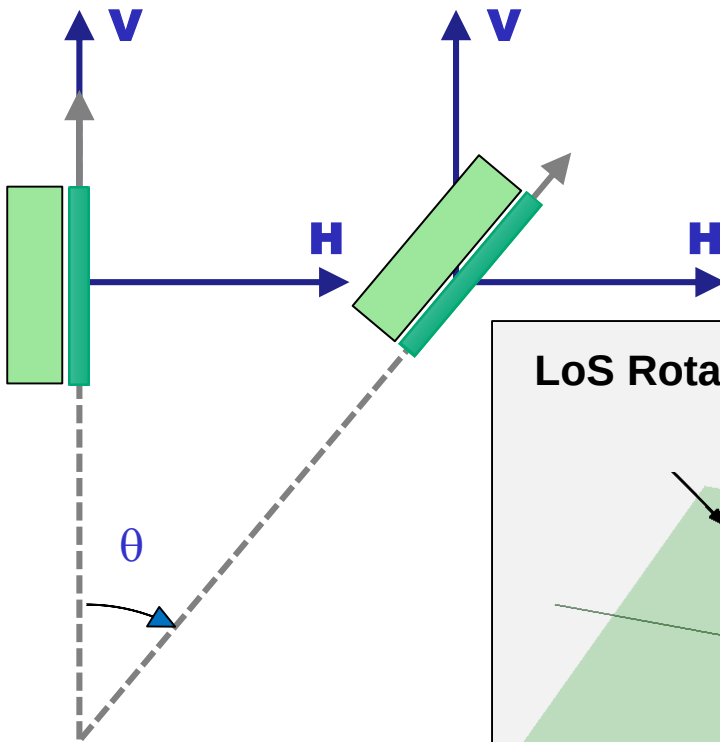
$$\begin{aligned} S_{HH}(\theta) &= S_{HH} \cos^2 \theta & + & S_{xx} \cos \theta \sin \theta & + & S_{VV} \sin^2 \theta \\ S_{xx}(\theta) &= -S_{HH} \cos \theta \sin \theta & + & S_{xx} (\cos^2 \theta - \sin^2 \theta) & + & S_{VV} \cos \theta \sin \theta \\ S_{VV}(\theta) &= S_{HH} \sin^2 \theta & - & S_{xx} \cos \theta \sin \theta & + & S_{VV} \cos^2 \theta \end{aligned}$$

$$S_{HV} = S_{VH} = S_{xx}$$

Line of Sight (LoS) Rotations: Dihedral Example

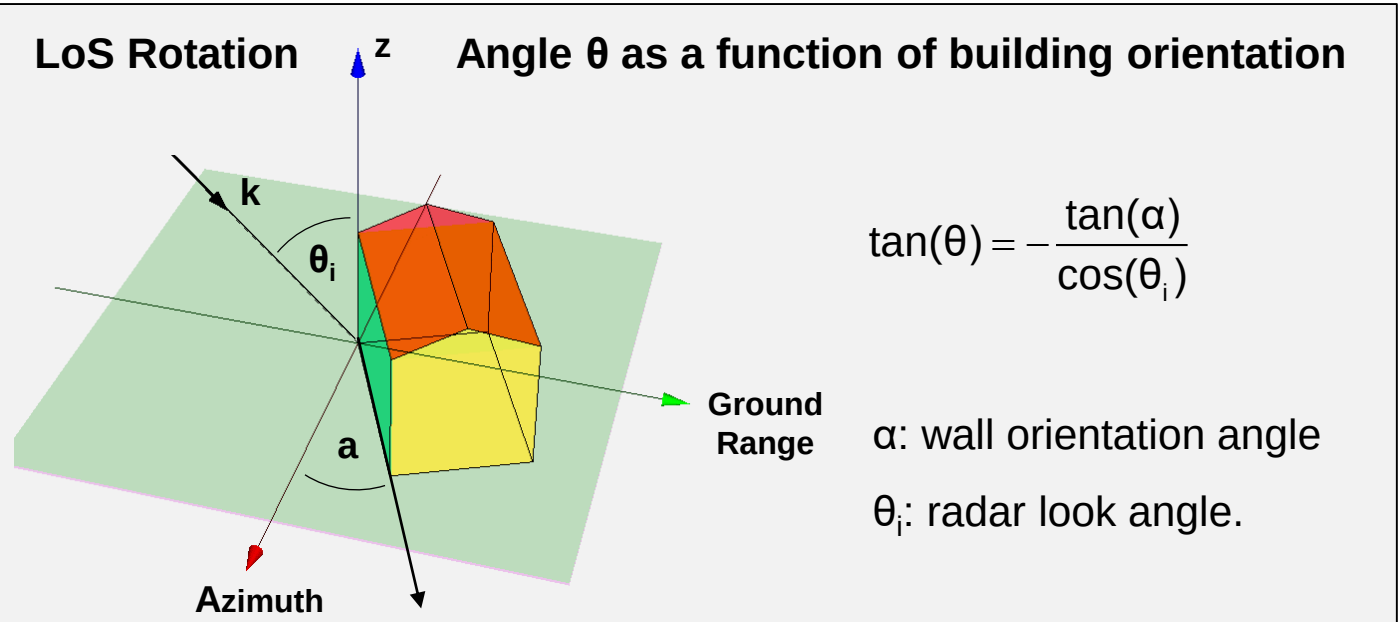
LOS Rotation of the Scatterer by θ = LOS Rotation of the Antenna by $-\theta$

$$[S] = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \xrightarrow{\text{Rotation}} [S(\theta)] = \begin{bmatrix} S_{HH}(\theta) & S_{HV}(\theta) \\ S_{VH}(\theta) & S_{VV}(\theta) \end{bmatrix} = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$



$$\begin{bmatrix} S_{HH}(\theta) & S_{HV}(\theta) \\ S_{VH}(\theta) & S_{VV}(\theta) \end{bmatrix} = \begin{bmatrix} \cos(2\theta) & \sin(2\theta) \\ -\sin(2\theta) & \cos(2\theta) \end{bmatrix}$$

LoS Rotation **Angle θ as a function of building orientation**



$$\tan(\theta) = -\frac{\tan(\alpha)}{\cos(\theta_i)}$$

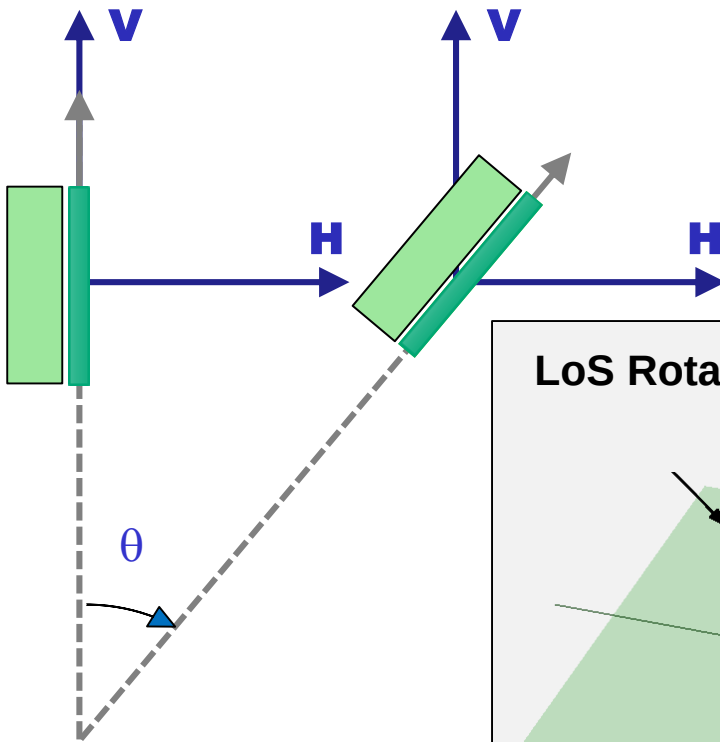
α : wall orientation angle

θ_i : radar look angle.

Line of Sight (LoS) Rotations: Dihedral Example

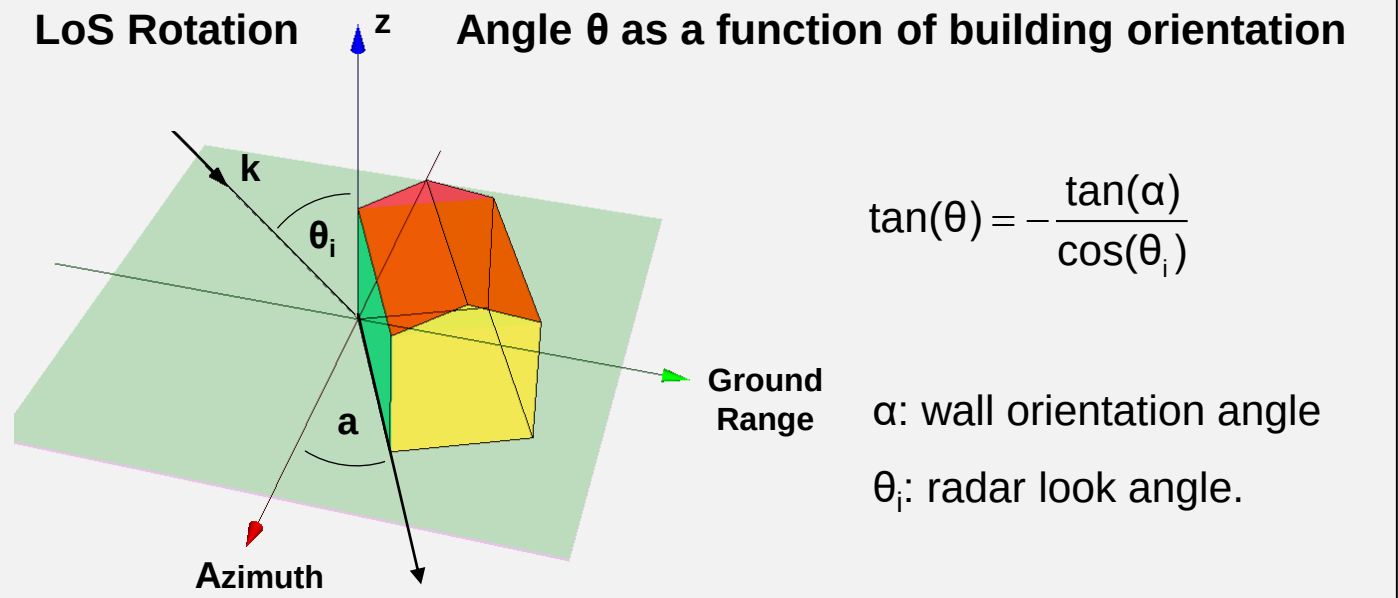
LOS Rotation of the Scatterer by θ = LOS Rotation of the Antenna by $-\theta$

$$[S] = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \xrightarrow{\text{Rotation}} [S(\theta)] = \begin{bmatrix} S_{HH}(\theta) & S_{HV}(\theta) \\ S_{VH}(\theta) & S_{VV}(\theta) \end{bmatrix} = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$



$$\begin{bmatrix} S_{HH}(\pi/4) & S_{HV}(\pi/4) \\ S_{VH}(\pi/4) & S_{VV}(\pi/4) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

LoS Rotation **Angle θ as a function of building orientation**





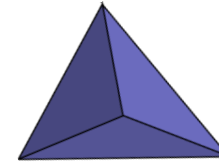
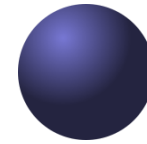


Pauli Decomposition

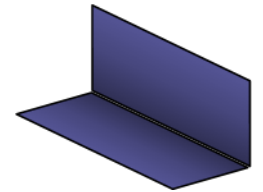
... decomposition of the scattering matrix into four scattering mechanisms:

$$[S] = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} = \begin{bmatrix} a+b & c-id \\ c+id & a-b \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + b \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} + c \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + d \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

- $[S_a] = a \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ **Single Scattering:** $a = S_{HH} + S_{VV}$

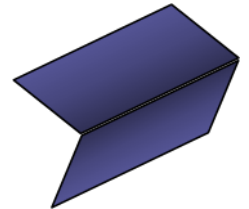


- $[S_b] = b \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ **Dihedral Scattering:** $b = S_{HH} - S_{VV}$



- $[S_c] = c \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ **Dihedral Scattering:** $c = S_{HV} + S_{VH}$

(... rotated by $\pi/4$ about the LOS – Line of Sight)

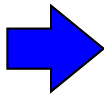


- $[S_d] = d \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$ **Helix Scattering:** $d = S_{HV} - S_{VH} = 0$ in backscattering !!!



In the monostatic case $[S]$ is decomposed into three orthogonal scattering mechanisms $[S_a]$, $[S_b]$, $[S_c]$.

Interpretation of Scattering Mechanisms

Scattering Matrix $[S] = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix}$  Pauli Scattering Vector $\vec{k}_{3P} = \frac{1}{\sqrt{2}} \begin{bmatrix} S_{HH} + S_{VV} \\ S_{HH} - S_{VV} \\ 2S_{XX} \end{bmatrix}$

Unitary Representation of $[S]$: $\vec{e} = \frac{1}{|\vec{k}_{3P}|} \vec{k}_{3P} = \frac{1}{\sqrt{2} |\vec{k}_{3P}|} \begin{bmatrix} S_{HH} + S_{VV} \\ S_{HH} - S_{VV} \\ 2S_{XX} \end{bmatrix} = \begin{bmatrix} \cos\alpha \exp(i\gamma) \\ \sin\alpha \cos\beta \exp(i\delta) \\ \sin\alpha \sin\beta \exp(i\epsilon) \end{bmatrix}$

Parameterisation of $\vec{e}$ in terms of five (scattering) angles: α , β , γ , δ , ϵ

where $|\vec{k}_{3P}| = TP = \frac{1}{2} |S_{HH} + S_{VV}|^2 + \frac{1}{2} |S_{HH} - S_{VV}|^2 + \frac{1}{2} |S_{XX} + S_{XX}|^2$ and $S_{XX} = S_{HV} = S_{VH}$

Point Reduction Theorem:

(Scattering) Phase Term

LOS rotation*

Scattering mechanism

$$\vec{e} = \begin{bmatrix} \cos\alpha \exp(i\varphi_1) \\ \sin\alpha \cos\beta \exp(i\varphi_2) \\ \sin\alpha \sin\beta \exp(i\varphi_3) \end{bmatrix} = \begin{bmatrix} \exp(i\varphi_1) & 0 & 0 \\ 0 & \exp(i\varphi_2) & 0 \\ 0 & 0 & \exp(i\varphi_3) \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\beta & -\sin\beta \\ 0 & \sin\beta & \cos\beta \end{bmatrix} \begin{bmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

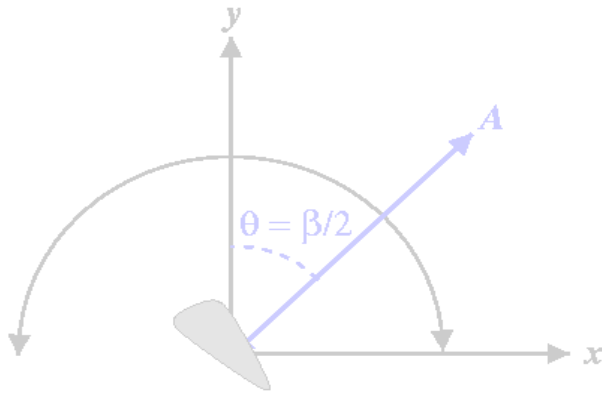
*) $\theta_{LOS} = \beta/2$

Interpretation of Scattering Mechanisms

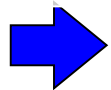
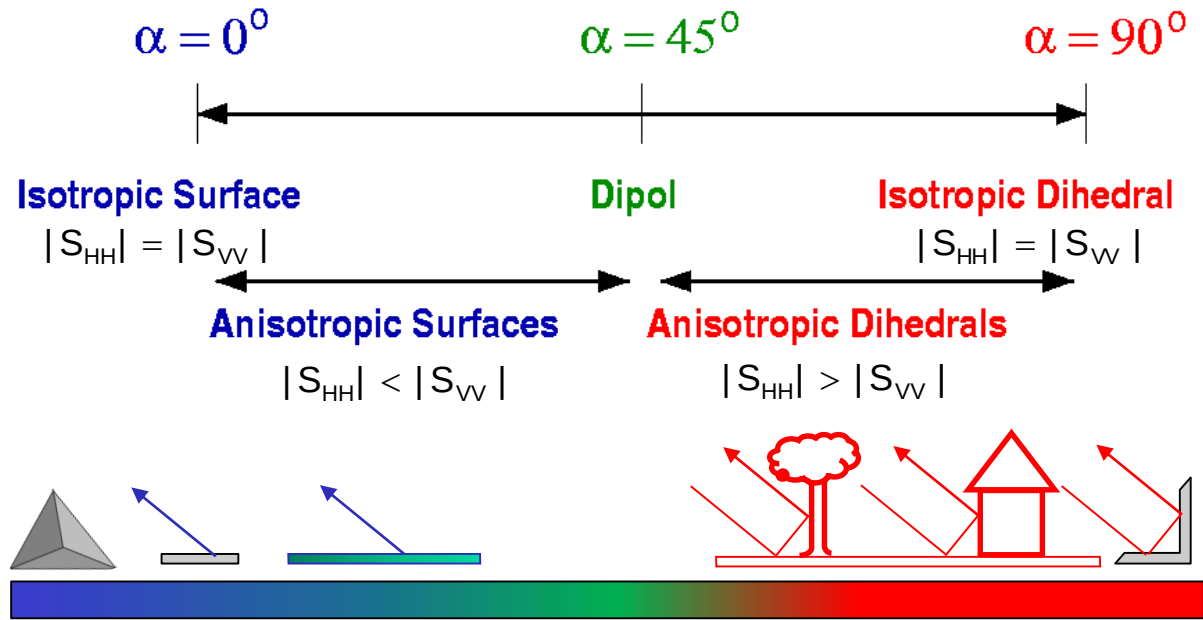
Unitary Representation of [S]:

$$\bar{\mathbf{e}} = \begin{bmatrix} \mathbf{e}_1 \\ \mathbf{e}_2 \\ \mathbf{e}_3 \end{bmatrix} = \frac{1}{|\bar{\mathbf{k}}_{3P}|} \bar{\mathbf{k}}_{3P} = \frac{1}{\sqrt{2} |\bar{\mathbf{k}}_{3P}|} \begin{bmatrix} S_{HH} + S_{VV} \\ S_{HH} - S_{VV} \\ 2S_{XX} \end{bmatrix} = \begin{bmatrix} \cos\alpha \exp(i\gamma) \\ \sin\alpha \cos\beta \exp(i\delta) \\ \sin\alpha \sin\beta \exp(i\varepsilon) \end{bmatrix} \quad \Rightarrow \quad \alpha = \arccos(|\mathbf{e}_1|) \in [0, 90^\circ]$$

LOS orientation



Scattering mechanism



Polarimetry is sensitive to orientation, structure and dielectric properties of scatteter(s).

Scattering Amplitudes / α -Angle



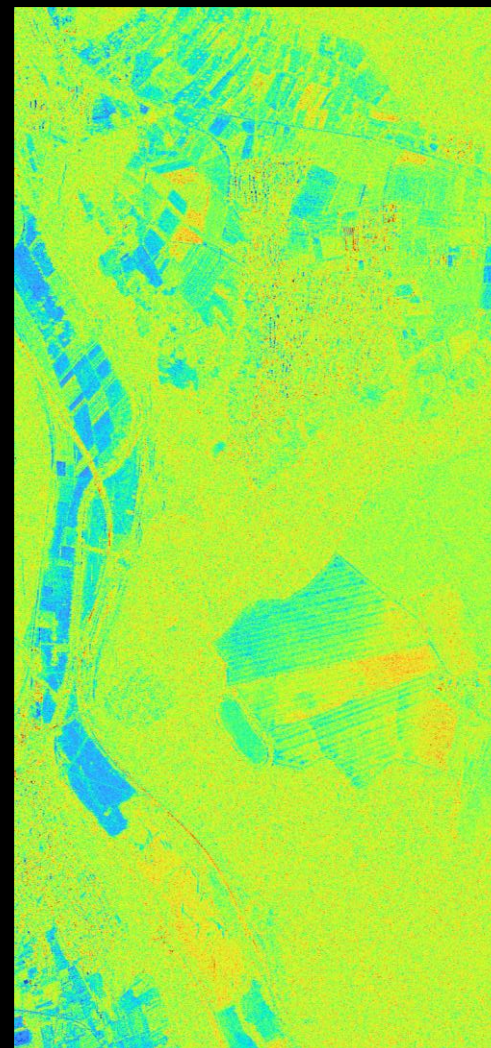
HH



HV



VV

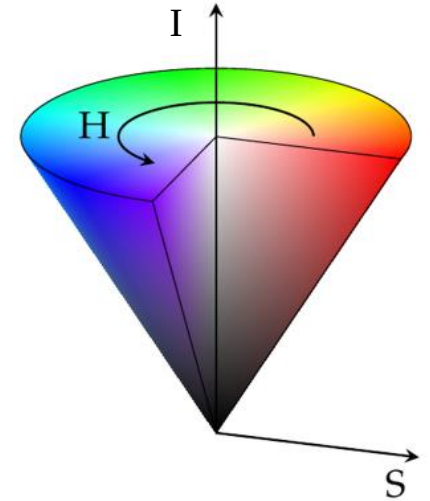


α -Angle

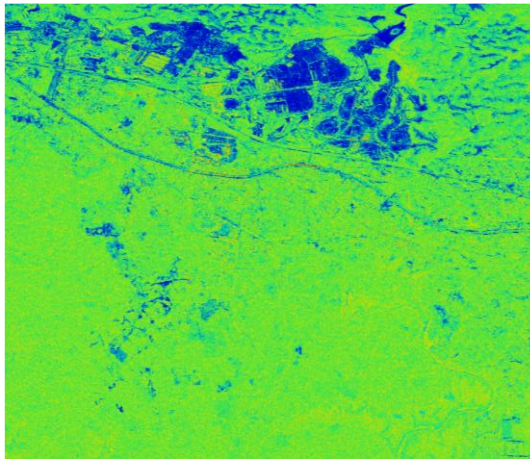
Hue Saturation Intensity (HSI) Color Representation

The HSI color model represents every color with three components: Hue (H), Saturation (S), Intensity (I).

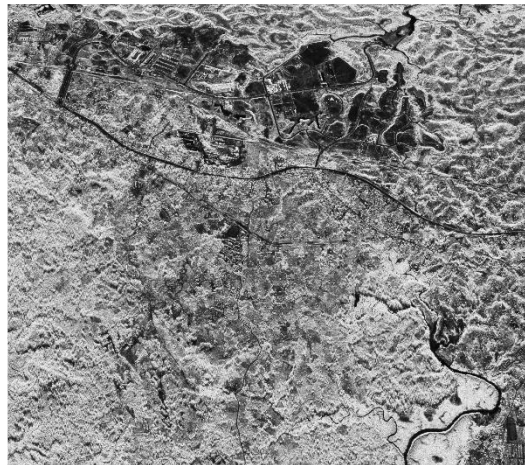
- The Hue component express the color itself in the form of an angle between $[0^\circ, 360^\circ]$. Red is at 0° , green at 120° and blue at 240° .
- The Saturation component express the dominance of the color (e.g. how much the color is diluted with white color). The range of the S is $[0,1]$.
- The Intensity component express how light or dark the color is range is between $[0,1]$. 0 means black, 1 means white.



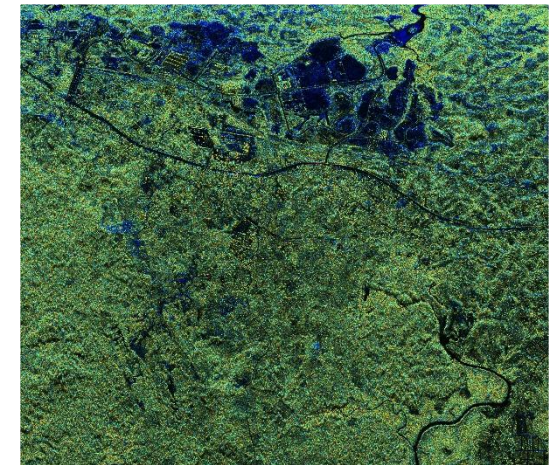
Polarimetric HSI Color Composite: $H=4 \cdot \text{Alpha angle}$, $S=1$, I : Total Power scaled from 0 to 1



Alpha angle



Total Power



Polarimetric HSI Color Composite

Scattering Amplitudes / HSI



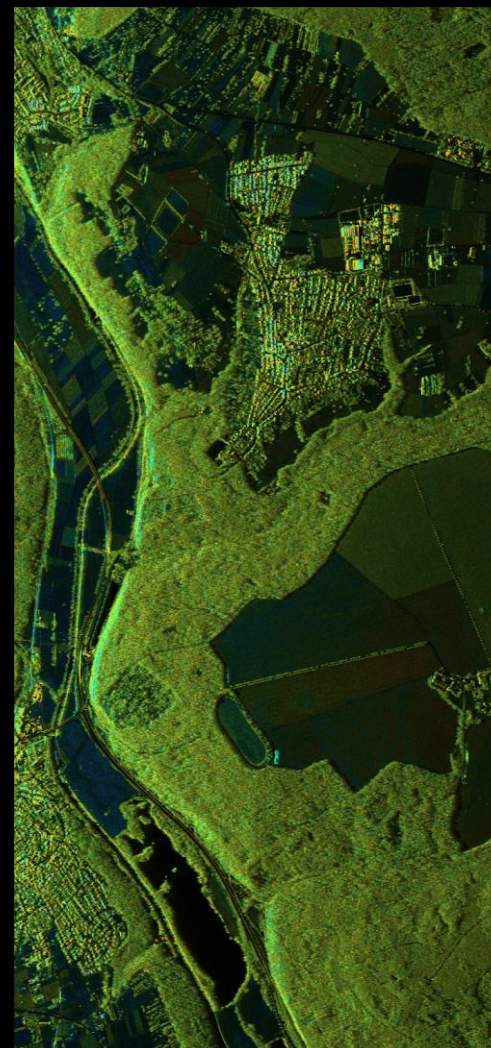
HH



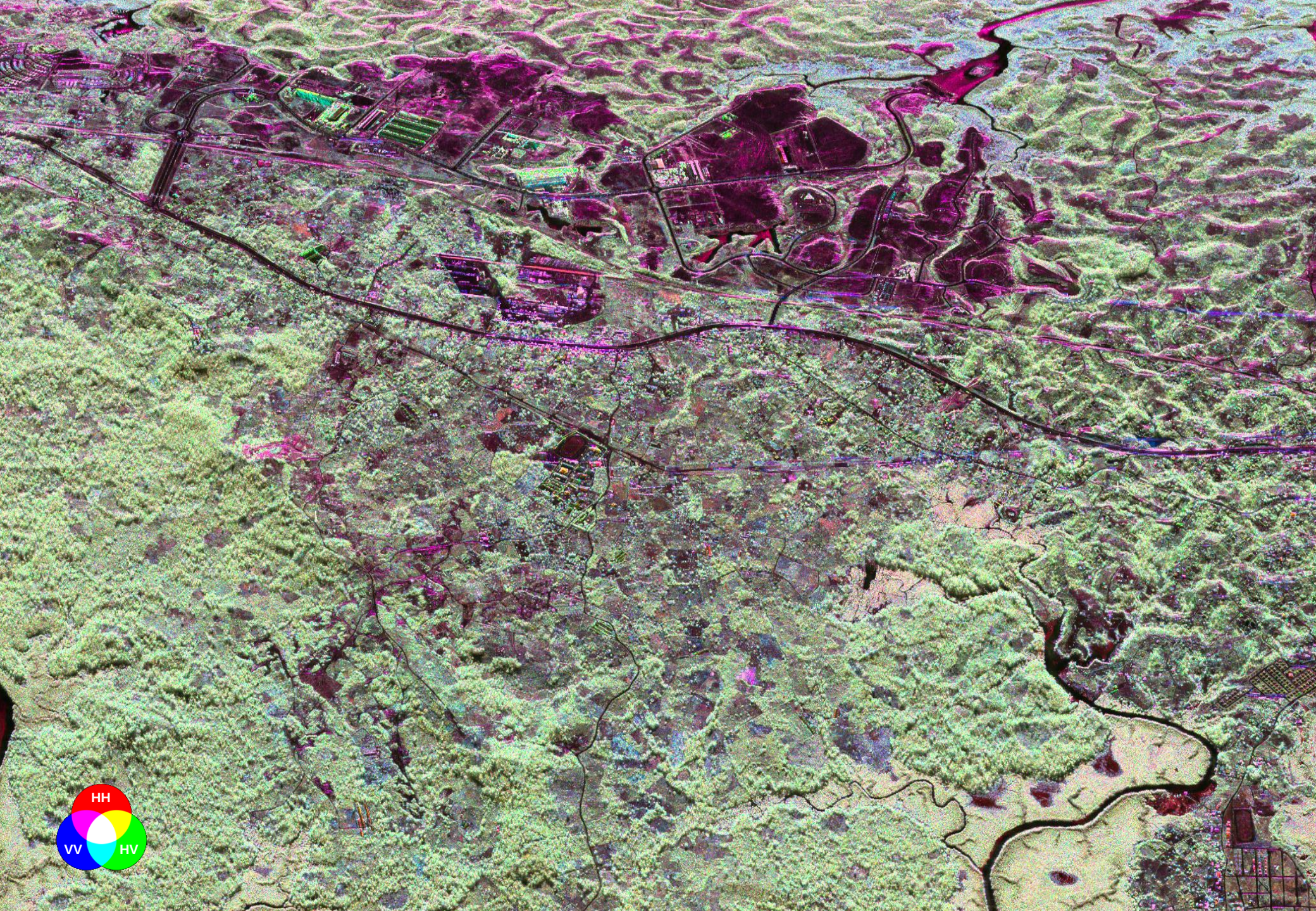
HV

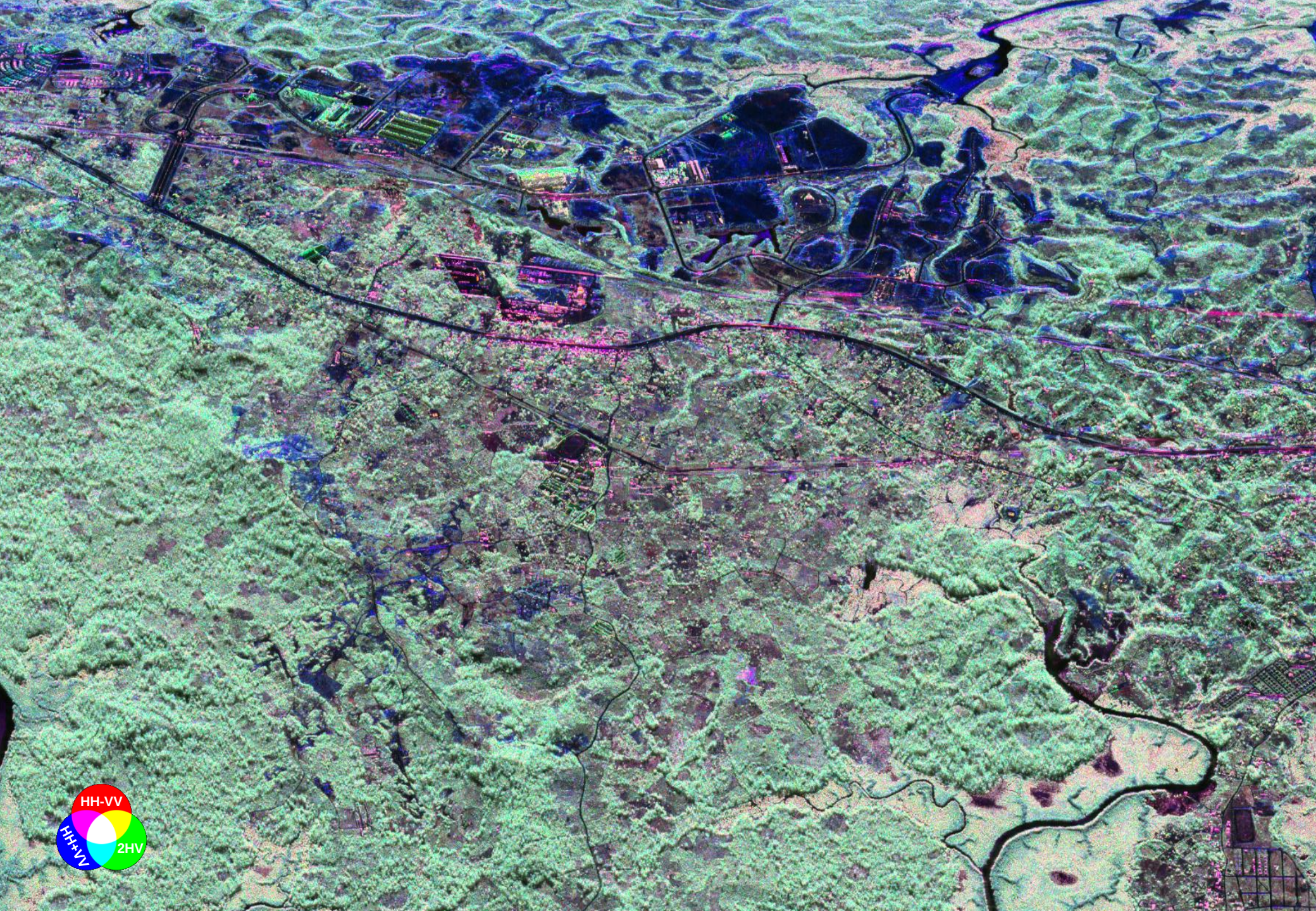


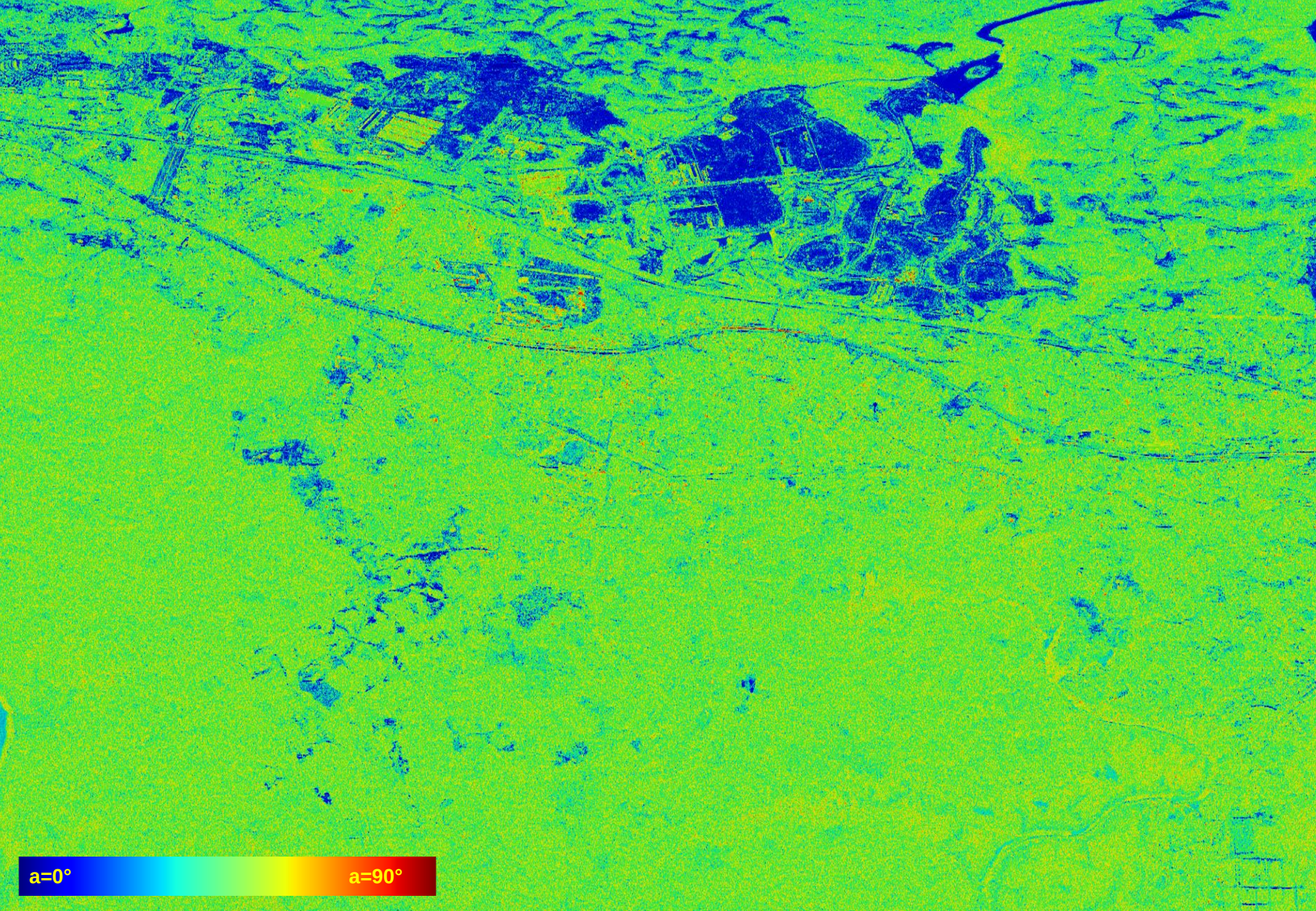
VV



HSI

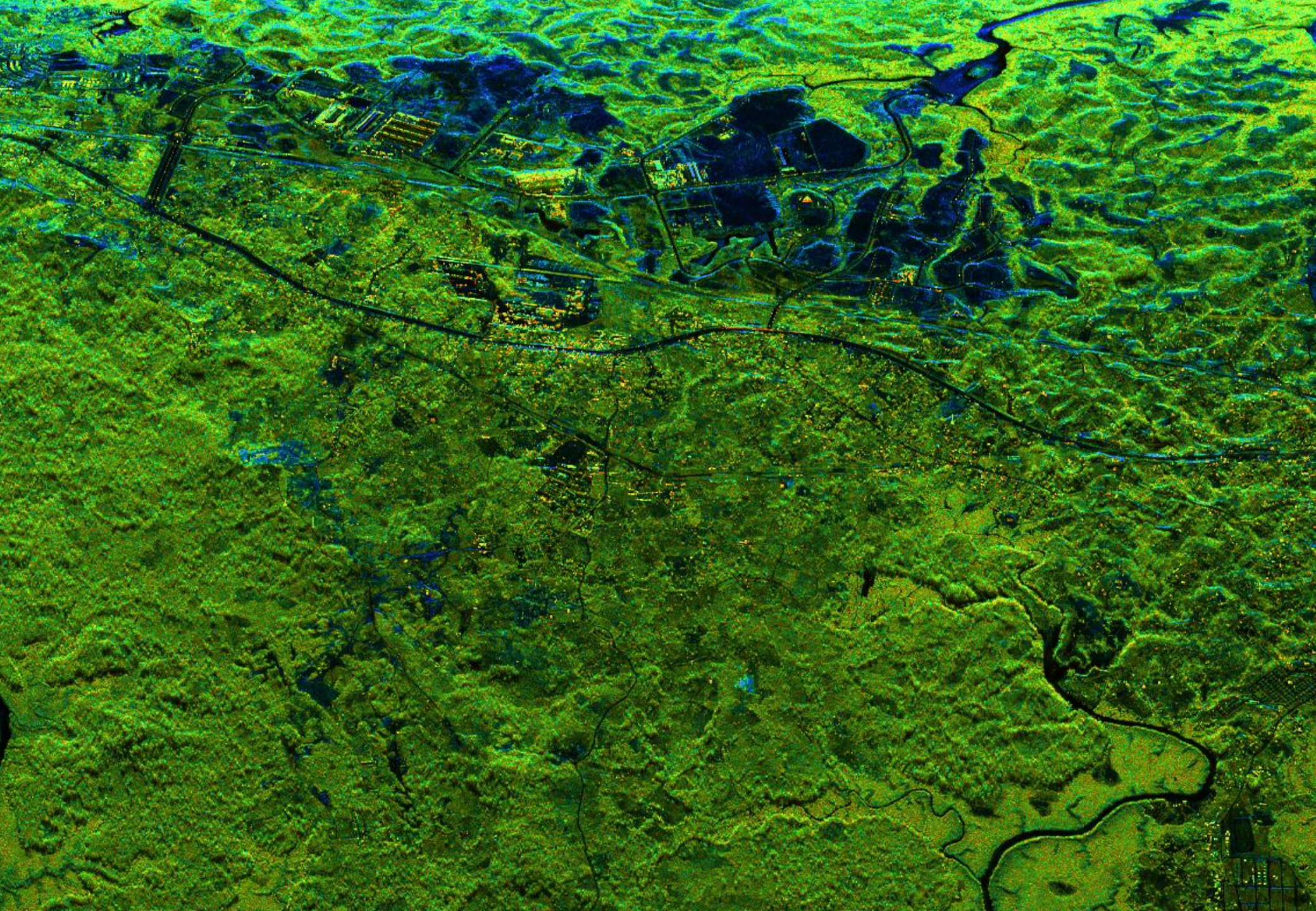






$a=0^\circ$

$a=90^\circ$



The Pol-InSAR Course

Synthetic Aperture Radar (SAR)

Part 6: SAR Polarimetry

Prepared by DLR-HR's Pol-InSAR Team

German Aerospace Center (DLR), Microwaves & Radar Institute (HR), Pol-InSAR Research Group

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